

A Students who has just entered the portals of colleges finds it difficult to understand the subject taught to him/her this difficulty is mainly due to understand standard references. While preparing this "THE PREPARATORY BOOK OF CELL AND MOLECULAR GENETICS" has been prepared to serve the special purpose of college students. This Preparatory book explains the basic Cell and Molecular genetics Structure and cytological technique in a simplified manner. This Preparatory book can be used for UG and PG Zoology students, who are looking for a quick revision before the Examination. This Preparatory book support to the zoology students to get confidence and also reference just before attending University Examination and other Competitive Examination. This Preparatory book has been written to Students and preparatory the field of Life sciences, No such book which cover suitable original picture with needed explanation of cell and molecular genetics techniques and Practice. This book is useful for the preparation of state Universities and College students integrated Certificate and Diploma courses field of Cell and Molecular genetics.



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THE PREPARATORY BOOK OF CELL AND MOLECULAR GENETICS

(Basic Concept of Cell & Molecular Genetics)



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CELL & MOLECULAR GENETICS

**THE PREPARATORY BOOK OF
CELL AND MOLECULAR GENETICS**

Second Edition-2022

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FOREWORD

The Preparatory Book adheres to the scope and sequence of most one major course nationwide. Also strive to make cell & molecular genetics as a discipline, interesting and accessible to students. In addition to a comprehensive coverage of core concepts and foundational research, incorporated features that draw learners into the discipline in meaningful ways.

This Preparatory Book provides a fundamental concept of cell & molecular genetics with a scope that is manageable for instructors and students alike with this; it is my pleasure to write a brief foreword to this wonderful Preparatory Book on “The Preparatory Book of Cell and Molecular Genetics”.

I am appreciating the Authors to chosen this book chapter on Basic Preparatory knowledge of Cell and molecular genetics. On the aspect of Cell Biology include number of books are available in the form of hand book, question bank for competitive examinations etc. except from this for the student community limited numbers of this like Preparatory book are available.

In UG or PG level lack of syllabus covered text books are available, this book might be useful to basic and advanced learner of cell & molecular genetics. Written of this Preparatory Manual contain simple language and easy to understand a basic learner. This Manual offers a valuable guide to the intellectual. This Preparatory book can help teachers develop a shared vision and understanding of basic concepts of cell & molecular genetics and Techniques of biology and its flexible uses across subjects and grade levels.

This Preparatory book has been created with several goals in mind, accessibility, customization, and student engagement all while encouraging students toward high levels of Academic scholarship. Instructors and students alike will find that this Preparatory book offers a strong introduction to cell & molecular genetics in an accessible format.



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PREFACE

A Student who has just entered the portals of colleges finds it difficult to understand the subject taught to him/her. This difficulty is mainly due to a lack of standard references. While preparing this “**THE PREPARATORY BOOK OF CELL AND MOLECULAR GENETICS**” has been prepared to serve the special purpose of college students.

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This book is useful for the preparation of state Universities and College students integrated Certificate and Diploma courses field of Cell and Molecular genetics.

Dr. S. KALAIMANI

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Words find no bounds to express my sincere thanks to all who made this Book to come true. my sincere thanks to Dr.C.Seeranganayahi, Principal, J.K.K.Nataraja college of arts and science, Komarapalayam, Namakkal Dt, for extending the facilities to undergo this manual work.

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My express my deep sense of feeling of my affectionate father late Mr.P.Subramani, sub-registrar, my mother late Mrs.s.veerammal, and my brother Mr.S.Velumani, Assistant Reg.dept, My wife Mrs.V.N.Padma, Assistant, CTO, My kids K.Deepthadhayazhini and K.Mithunmonish to whom i owe everything for their dedicated efforts and loving encouragement to complete this book successfully.

Last but not least i wish to extend my sincere thanks to all my friends and relatives for their co-operation and support in this manual to make success.

Dr. S. KALAIMANI

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Chapter 1

History of Cell Biology

- 1665- Robert hook-term of cell
- 1831- Robert brown –discovered Nucleus
- Cytology started beginning 17th century
- Microscope first used by Hook, Grew & Malpighi
- Cell theory- two German scientist Scheliden and Schwann.
- New cells originate from the preexisting cells –Robert Virchow 1858.
- Cellular totipotency- ability of a somatic cell of a plant to produce new complete plant.
- Cellular totipotency in carrot culture – Steward et al.
- Growth and synthesis of protoplasm- control by nucleus.
- Cytoplasm otherwise called – Cytosol, Mesoplasm, Cytoplasm.
- Cytoplasm word given by Kolliker.
- Protoplasm – cell wall – Nucleus
- Cytoplasm protoplasm - Nucleoplasm.
- Protoplasm – reversible crystal colloidal solution.
- Cytoplasm coagulates with acid /bases or heat 60o c.
- Smallest cell – PPLO
- Kart Von Baer- Discovered mammalian Ovum
- R. Brown- Brownian movement and Nucleus
- Felix Dyjardin- described Protoplasm
- H.Von mahl- Described cell division
- M.j schleiden - cell theory -Nucleoli-T Schwan – cell theory in animal
- J.E Purkinje – Named protoplasm
- Robert Remark- Described amitotic cell division in RBC
- A. Donne – Studied Spermatozoa
- H.Von mohl- Cytoplasm
- Kolliker- Mitochondria in Muscle
- R. Virchow – Cell are primary site of diseases
- Schultza- Protoplasm as Physical basis of life
- Walderyer- reported chromosome (Haemotoxylin)
- G. Mendel- Fundamental Principle of genetics.
- L.Gst eore- Discovered Golgi complex
- W.His – Microtome
- F.Miescher- Discovered NA (Nucleoprotein)

CELL & MOLECULAR GENETICS

- Cyclosis helps Formation of pseudopodia in amoeba, Movements of cell organelles of cell inclusion.
- Cell- Robert Hook -Cell- basic unit of structure and function of the cell.
- “Micrographia” –Robert hook.
- Karyoplasm=Nucleoplasm
- Cytoplasm inclusion -Glycogen granules, Fat droplets
- Micro fibrils (Actin and Myosin)- contractile organelles
- Ribosome (protein factory) otherwise called Microsomes.
- RER(Rough endoplasmic reticulum)-site for protein synthesis
- SER(smooth endoplasmic reticulum) site for lipid synthesis
- Lysosome – Digestion centre of cells.
- Mitochondria-respiratory centre of the cell
- Centrioles- cell division
- Golgi apparatus –secretion of enzyme and hormone.
- Plasma membrane otherwise called Plasmalemma .
- H.Fol- Described Spindle & Astral rays
- Stras burger- Described Chromosome
- L.Pasteur- Fermentation Process.
- O.Hertwi- Fertilization occur union of sperm & eggs
- W. Flemmins – term chromatin Spilicity of Chomosomes
- C.G Balbiani- Giant chromosome(salivary gland)
- W. Flamming – Staining term of mitosis
- Stras burer- Described mitosis in plays term cytoplasm & Nucloplasm
- W Pfitzer- Chromosome contain granules on the chromosome.
- Metschnioff – Phagocytosis in cell
- R. Altman- Stained Mitochondria &role of respiratory
- E. Van Beneden- Discovered centriols
- T Boveri- isolated Mitochondria
- Kolliker- 1857 by physiologist Albert von Kolliker -Mitochondria
- T.Boveri - Spermatogenesis & organogenesis
- C. Benda- named Mitochondria
- C.Golgi – Described in a complex in nerve cell

Chapter 2

Permanent Slide Preparation

Permanent slide preparation of Obelia- colon, Pharyngeal and septal nephridium of earthworm, Parapodia of Nereis and Heteronereis, Gill, Radula and osphradium of Pila, Salivary gland, mouth parts and Trachea of cockroach, Gills lamina of Unio, Statocyst and hastate plate of prawn.

INTRODUCTION:

A microscope slide is a thin flat piece of glass, typically 75 by 26 mm (3 by 1 inch) and about 1 mm thick, used to hold objects for examination under a microscope. Typically the object is placed or secured ("mounted") on the slide, and then both are inserted together in the microscope for viewing.

This arrangement allows several slide-mounted objects to be quickly inserted and removed from the microscope, labeled, transported, and stored in appropriate slide cases or folders. Microscope slides are often used together with a cover slip or cover glass, a smaller and thinner sheet of glass that is placed over the specimen. Slides are held in place on the microscope's stage by slide clips, slide clamps or a cross-table which is used to achieve precise, remote movement of the slide upon the microscope's stage.

The following points highlight the seven main processes involved in preparation of permanent slides.

The Processes Are:

1. Killing
2. Fixing and Hardening
3. Staining
4. Dehydration
5. Clearing
6. Mounting
7. Labeling.



Figure 1: ADVANCED MICROSCOPE

2.1. METHOD OF MICROSCOPIC SLIDE PREPARATIONS

For Microscopic studies the specific material (tissue organs or small organism) is mounted on a glass slide.

There are two methods of mounting the material on slide.

1. Temporary Mounting
2. Permanent Mounting

1. Temporary Mounting:

The temporary mount is prepared either in glycerin, water or normal saline. The material is first washed in tap, then stained and differentiated. Drop of mounting medium (glycerin and water) is placed on centre of the slide. The material is then transferred into that drop. It is then covered neatly with a cover slip. The excess of glycerin or water is absorbed by piece of blotting paper. Mount prepared by this method can be used for study only for few hours, after which material loses its original form due to diffusion and other post mortem changes.

2. Permanent Mounting:

The study of microorganisms, smaller animals and histological studies of tissues, an elaborate technique is employed for making their permanent preparations. These smaller objects are mounted in balsam on a slide. There is a series of processes by which a living organism or its tissue is made fit for microscopic examination in a permanent state. The utility of permanent preparation is that the animal cell or tissue remains as such without undergoing major changes.

2.2. THE PERMANENT PREPARATION INCLUDES:

1. Killing and Narcotisation
2. Fixing
3. Washing
4. Staining
5. De-staining or removal of excess of stain.
6. Clearing or de-alcoholization.
7. Mounting on slide.

1. Killing and Narcotisation:

The first step in permanent preparation is killing instantaneously in order to prevent the change in form of the object as it has in living condition and immediately fixing the object. Sometimes killing is preceded by Narcotization. The narcotics used are chloroform, menthol, ether, alcohol, acetone, etc. the purpose of narcotization and killing is important as to have the same form and chemically constructed tissue or organisms as it had during its lifetime. In certain cases, for smaller animals killing is heating done by the slide.

2. Fixation:

Fixing is done with various fixative agents for histological elements. Fixative is essential in every type of microscopic preparations, either for sections or for whole mounts and also in larger specimens.

Function of Fixation:

- The tissues become hard and hardening resists further post-mortem changes.
- Fixative agent coagulates and renders insoluble elements of tissues which are dissolved in further processing.
- The fixative agent renders insoluble the various constituent elements of cells, alters their refractive indices and thus makes them optically differentiated under the microscope. Because of Brownian motion there is no possibility of material but we must bear in mind that fixed details are the coagulation artifact of the living structures.
- Various fixative agents generally used are absolute alcohol, 90% alcohol plus glycerin, picric acid, corrosive sublimate, formal, osmium tetroxide and nitric acid with or without water.
- Depending upon the material, corrosive sublimate or alcohol Carnoy's fluid for cytological studies and other fixative for histochemical studies.

3. Washing:

Washing is essential as by this process the uncombined and excess of fixative agent is removed. The presence of fixative agent in tissues or cells will inhibit good staining. The washing agent depends upon the type of fixative agent used. As alcoholic picric acid in water is removed by 70% alcohol. Formal and corrosive sublimate are washed with water distillate. Sublimate is washed in alcohol.

4. Staining:

The tissue or cell components are stained in various dyes. The dye makes the tissues distinct in its histological sphere. The various dyes are Orange G, Bordeaux red, Sudan's Congo red, Alizarins oxyquinoine, methylene blue, neutral red, borax carmine, heamatoxylin, epicroindigo carmine, eosin and Gower's carmine.

Mainly two kinds of stains are used.

1. Nuclear stains. Stains the nuclear parts of the cells, such as Delafield's or Erhlich's haematoxylin.
2. Cytoplasmic stains such as borax carmine, picro-indigo carmine, Gower's carmine and eosin, etc., which stain cytoplasm. For general staining borax carmine is used aqueous stains are prepared in water whereas alcoholic stains are prepared in alcohol. When a single stain is used the process is called as simple or single staining. In some cases two stains, i.e., Nuclear and Cytoplasmic are used is called as double staining. Generally single stain is used for whole mounts but for protozoan's etc., both cytoplasmic and nuclear stains are used.

5. De staining:

The removal of excess of stain is called as de staining or differentiation. De-staining agents are acid alcohol or acid water. The acid alcohol is used with alcoholic stains while acid water is used with aqueous stains.

6. Dehydration:

This process is meant for removal of water from the tissues. The dehydration prevents putrefaction or decaying and maintains the same shape and size of tissues or cells. The moisture or water in tissues absorbs various germs of destructive nature so that the tissue may be destroyed, hence the passing the mounting material through various grades of alcohol, such as 30, 50, 70, 90 and 100% alcohols. The tissue is soaked in gradually increasing strengths of alcohol. The lower grads prepared either from 90% or absolute alcohol. The dehydration is carried out in corked or glass-stopper tubes.

7. De-Alcoholization or Clearing:

After dehydration, transparency in tissues is obtained by treating with a clearing agent, which removes alcohol and makes the tissue clear and transparent. The clearing agents are wood oil, clove oil, xylol and benzol, etc. Xylol is most commonly employed and it makes the tissues hard and brittle. Clove oil is a superior clearing agent especially in the whole mounts. It also possesses higher index of refraction than balsam mounting media.

8. Mounting:

Mounting forms the end of permanent preparation the choice of mounting media is not much but they should have the same refractive index as that of the cleared tissue. The refractive index of such a stained, dehydrated and cleared cells is 1.54. Canada balsam or D.P.X has almost the same refractive index. Mounting is an easy process.

The tissue is kept over glass slide in a drop of balsam and cover-slip is lowered slightly. After mounting, the excess of balsam on the slide, as generally happens with beginners, should be removed with cotton soaked with the balsam has dried.

For much better finishing the edge of the cover-glass may be ringed with cement such as gold seal or a varnish. The air bubbles present in balsam under cover-glass should be removed by gentle heating. During all the chemical bathing of tissues, two changes of each reagent are necessary. The time of keeping tissue in various reagents may vary from 5 to 15 minute.

2.3. PRECAUTIONS AND INSTRUCTIONS

1. The articles, such as slide, cover slips and instruments should be perfectly cleaned.
2. The working place should be kept in order.
3. During dehydration, the tissues should be kept in tightly closed cork or glass stopper tubes. The opened tube will spoil material by absorbing moisture from atmosphere. Even breathing closely with dehydrating tube is undesirable.
4. The change of solution is done very quickly, reducing time of exposure to atmosphere to minimum.
5. The chemicals used once should not be reutilized.
6. The Canada balsam used should be clean, dust-free and not viscous.



Figure 2: MICROSCOPIC SLIDE

2.4. DIFFERENCE BETWEEN ANIMAL CELL AND PLANT CELL:

1. **ANIMAL CELLS:** Animals are made up of millions of cells. Animal cells have an irregular structure and are made up of four key parts:

1. Nucleus – This contains genetic material (DNA), and controls the cell's activity.
2. Cell Membrane – A flexible layer that surrounds the cell and controls the substances that enters and exit.
3. Cytoplasm – A jelly-like substance where the chemical reactions happen.
4. Mitochondria – This is where energy is released from the food molecules.

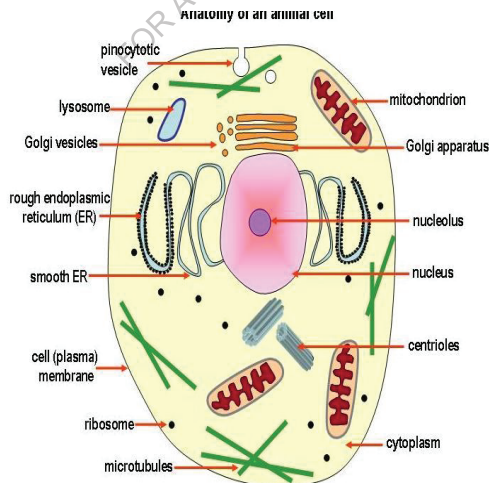


Figure 3: ANIMAL CELL

2. PLANT CELL:

Plants are also made up of millions of cells. Plant cells have a nucleus, cell membrane, cytoplasm and mitochondria, but they also contain the following structures:

1. **Cell Wall** – A hard layer outside the cell membrane, containing cellulose to provide strength to the plant.
2. **Vacuole** – A space inside the cell that is used to store substances and help the cell keep its shape.
3. **Chloroplasts** – Structures that contain the green pigment chlorophyll, which are a key part of photosynthesis.

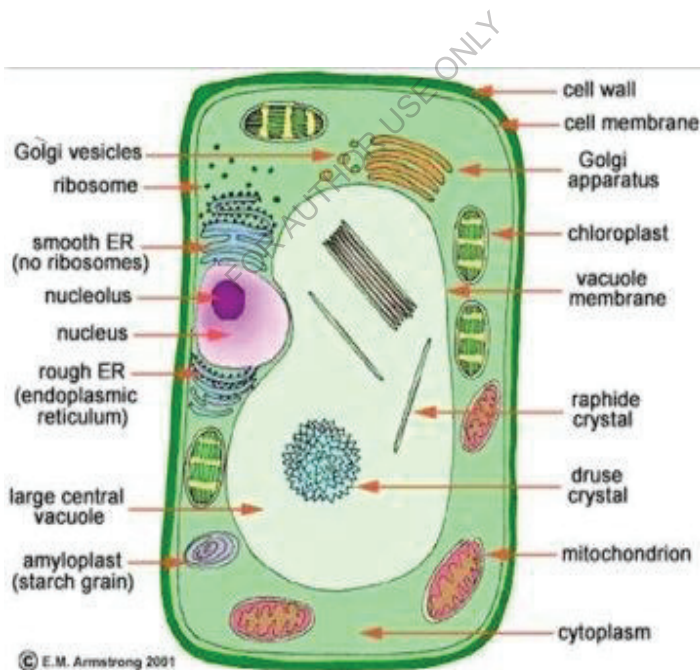


Figure 4: PLANT CELL

Chapter 3

Cell Theory**History of cell theory**

- Charles Darwin theory of evolution, the theory of the gene of modern biology.
- PPLO-Pleuropneumonia like organism e.g. Mycoplasma gallisepticum- bacterium (0.1-0.3 μ m).
- Viruses- biological puzzles.(living –non living)
- First give the evidence Iwanosky (1891)
- First determine composition of virus by “Schlesinger” (1933).
- TMV-1935 –stanly (crystallized TMV).
- Infect virus unit or infectious particles called virion.
- Non living character of viruses – Absence of its own enzyme system absence of growth lack of any response to stimulus they can be crystallized.
- Living character- Ability to multiple in host cell property of mutability, recombination inheritability.
- Prokaryotes- Bacteria, blue green algae, mycoplasmaor, PPLO, Spirochaetes and rickettiae
- Cell first to raise Evolutionary (3x106years old).
- Eukaryotes arise a billion years.
- Cell- Robert Hook, Cell- basic unit of structure and function.
- The book of ‘Micrographia’ –Robert hook.
- Prokaryotic genetic materials contain DNA but no histone protein.
- 1957 “Dougherty” tem prokaryotes and Eukaryotes.
- Blue green algae now called cyanobacteria
- Tubular network of membrane organelle “Endoplasmic reticulum”.
- Mesokaryotes – genetic material surrounded ,nuclear membrane (Eukaryotes), Histone present, absent (prokaryotes)
- Cell – Second law of thermodynamics every living organism has tendency to increase –Entropy
- Cells –10-100 μ m, smallest cell PPLO e.g. mycoplasma gallisepticum bacteria (0.1-0.3 μ m),Cells shorted length e.g. Dialister pneumosinates (0.15-0.3 μ m).
- Longer unicellular plant e.g. Acetabularia (upto 10cm).
- umbrella plant algae e.g. Acetabularia,Longer plant cell- fibrous of ramie e.g. Boehmeria ulrea (upto55cm).
- Largest cell - (plant) e.g. Cycus ovule.
- Largest cell - (animal) e.g. Ostrich egg (11x170cm).

- Largest cell –e.g. Nerve cell up to 90cm long.

3.1 CELL THEORY

Cell theory- German scientist Scheliden and Schwann

1. All living organisms are composed of one or more cells.
2. The cell is the basic unit of structure and organization in organisms.
3. Cells arise from pre-existing cells.

3.2.DIFFERENCE BETWEEN PROKARYOTES AND EUKARYOTES BASED ON THEIR STRUCTURE AND FUNCTION.

PROKARYOTES	EUKARYOTES
1. Bacteria blue green algae (Cyanobacteria)Mycoplasma (PPLO)	1. Algae, protista and, metazoans plants
2. Photosynthesis takes place carboxylsomes or lamellae	2. Photosynthesis takes place chloroplast only.
3. Size 0.1-5 μm	3. Size 5-100 μm
4. Cell wall muramic acid (Peptidoglycon)	4. Cell wall made up of cellulose, chitin.
5. Mucopolysaccharides capsule - present	5. Capsule is absent.
6. Mesosome control cellular respiration and cell division	6. Mesosome absent Respiratory enzyme found in cytoplasm & mitochondria
7. Naked chromatin network otherwise called Nucleoid or genophore.	7. Nucleus is present.
8. Multiple Nucleoid	8. Mononucleoid single Nucleus
9. DNA(circular)Naked	9. DNA Linear with Histone (Histone protein absent in mitochondria & plastids DNA).
10. Poly cistronic DNA	10. Mono cistronic DNA
11. One kind of RNA polymerase is present	11. Three kinds of polymerase rRNA, mRNA, tRNA
12. Transcription is occur in cytoplasm	12. Transcription is occur in Nucleus Translation occur in cytoplasm
13. Circular extra chromosomal DNA Plasmid or Episome.	13. Absent.

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14. Ribosome 70s(50s+30s)	11. Ribosome 80s (60s+40s)
15. Protein synthesis inhibited by chloramphenicol	12. Protein synthesis inhibited by Cyclohexamide
16. Flagella (flagellin)	13. Cilia & Flagella -9+2 arrangement ,microtubules made up of tubulin
17. Microfilaments Actin & myosin absent	14. Microfilaments Actin & myosin present
18. Gas vacuole are present	15. Sap vacuole are common in plant
19. Cytoplasm no streaming movements	16. Present streaming movements or cyclosis common in cytoplasm
20. Respiratory enzyme associated with pm& mesosome	17. Respiratory enzyme present in cytoplasm and mitochondria
21. Cardiolipin lipid is present	18. Cardiolipin lipid is absent
22. Endocytosis and Exocytosis are absent	19. Endocytosis and Exocytosis are present
23. Divide by Amitosis or simple fission	20. Mitosis & Meiosis cell division
24. Centriole is absent	21. Centriole present in animal and absent in plant cell
25. No spindle formation during division	22. spindle formation occurs during division

Chapter 4

Cell Organelles

4.1. PROTOPLASM:

- Term protoplasm “purkinji”
- Protoplasm=Bioplasm.
- Protoplasm is a polyphasic, colloidal.
- Protoplasm generates “Energy of life”(chemical energy- biological energy).
- Protoplasm ‘physical basis of life’ by J.Huxley.

COMPOSITION OF PROTOPLASM:

- water-75-85%, protein-10-20%,
- Lipid-2-3%
- Carbohydrates 1%
- Inorganic materials-1%.
- Major elements of protoplasm-C-20%, O-62%, H-10%, N -3%,
- Remaining 5%- Ca, P, Cl, S, K, Na, Mg, I, Fe.
- Protoplasm is crystalloid-colloid mixture.
- Colloidal theory of protoplasm by Fischer (1894) & Hardy (1899).
- Protoplasm is “Heterogeneous” in nature.
- Protoplasm shows intracellular movements called streaming movements. (Cyclosis).
- Protoplasm rotation clockwise /anticlockwise around vacuole-leaf cells of hydrilla & vallisneria.
- Protoplasm circulates around vacuoles e.g. stamina hairs of tradescandia.
- Eukaryotic cyclosis occurs based on (Viscosity of cytoplasm of temperature).
- Protoplasm exhibit Brownian movement & Tyndale effect.
- Protoplasm coagulates heated up to 60°C.
- Protoplasm has specific gravity – more than 1
- Protoplasm pH-slightly alkaline.
- Response of protoplasm to stimulus is called irritability.

4.2. BIOMEMBRANES

- Semi permeable.
- Covered around cell and organelles.
- It separated from external environment.
- Can be extended folded and repaired quickly.
- Play vital function in a living cell.
- Biomembrane covered the cell is called “cell membrane”- Nageli & Cramer.
- Biomembrane are plasma lemma – plowe 1931.
- Biomembrane covers the organelles called sub cellular membrane.
- Biomembrane are tripartite or trilaminar.

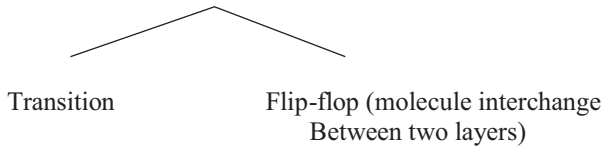
A. Composition of Biomembrane

1. Protein (59-75%) -Fibrous or globular-structural, carrier, receptor, enzymatic, glycoprotein.
2. Lipids (20-40%)—phospholipids, sterol, glycolipids, sphingolipid.
3. Carbohydrates (1-5%)—Hexose, hexamine, sialic acid.

B. Model of Bio membrane

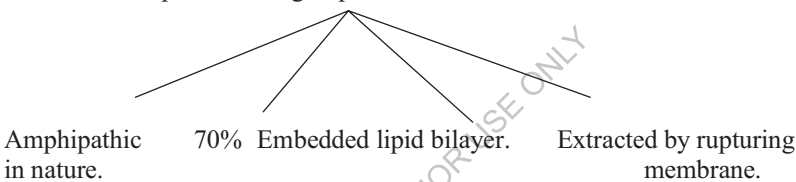
- Sandwich model P-L-P model (1935) -Danielli & Davson model (1935).
- Sandwich model are trilaminar lipoprotein
- α protein (1nm)
- Glycerol (polar)-hydrophobic
- Fatty acid (non polar)-Hydrophilic
- Between α protein & Fatty acid has hydrogen bond, Electrostatic force, ionic linkage.8nm or 7-10 Å°.
- In between polar and non polar lipid molecule bonded by Van der wals forces.
- Unit membrane-Robertson model (1959)- 75-100 Å° 7.5nm.
- Unit membrane present in Nucleus & mitochondria.
- Unit membrane not explains membrane stability & permeability.
- Fluid mosaic model / Singer & Nicholson Model) 1972) ---- Most accepted, trilaminar, Lipo protein model, glycerol and fatty acid gives membrane fluidity.

- Phospholipids contain two types of movements



(Molecular change their position)

- Intrinsic protein & extrinsic protein are called mosaic pattern.(Protein form icebergs in sea of lipids)
- Extrinsic protein or Peripheral protein --30% ,easily removed, contain Glycolipid/glycol protein, provide structural & functional specificity to the membrane.
- Example of Extrinsic protein --- ATP ase, spectrin, acetyl cholinesterase.
- Intrinsic protein /Integral protein



- Example of Intrinsic protein/Integral protein--- Cytochrome oxidase, rhodopsin.
- Some intrinsic protein inside the lipid bilayer called tunnel protein/transmembrane protein.

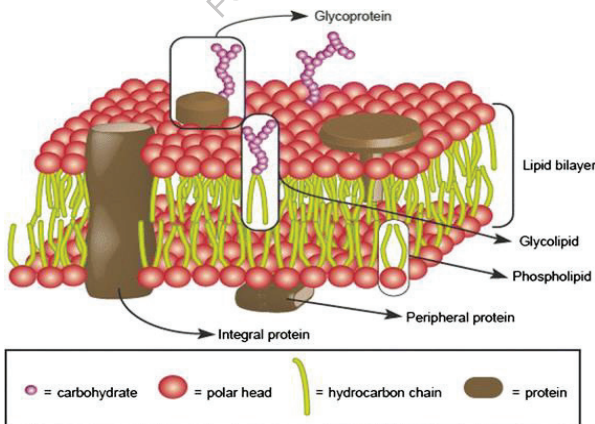
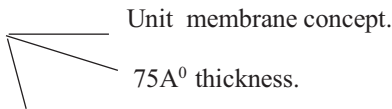


Figure 1: FLUID MOSAIC MODEL OF PLASMA MEMBRANE

PLASMA MEMBRANE:

- Thin, film –like pliable membrane called Plasma membrane or Plasma lemma.
- Eukaryotic cell (PM- Sub cellular membrane).
- PM is thin layer of lipids overtone (1902).
- 1935- Danielle and Davison, Bimolecular layer of Phospholipids ($35\text{\AA}$), Tri lamellar Sandwiched between two layers of Protein ($20\text{\AA}$).
- Total thickness of PM $20\text{\AA} + 35\text{\AA} + 20\text{\AA} = 75\text{\AA}$
- All biological membrane – Trilaminar structure.

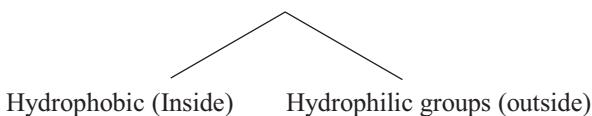
1959 Robertson



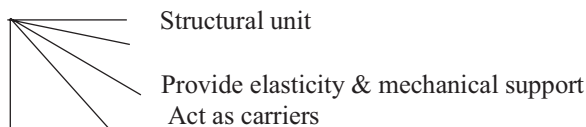
Trilaminar appearance.

1972-singer and Nicholson – Fluid Mosaic model
(Latest most widely accepted model)

- Protein icebergs in a sea of lipids described cell membrane by singer and Nicholson
- PM contain lipid 32%, Protein 42%, Carbohydrate, Water%
- Polar heads – lipid molecule-Glycolipid
- Protein exposed outer surface-glycoprotein
- Sugar Protein of Glycolipid- recognition mechanism
- Sugar- recognition sites-bind each other causes cell to cell adhesive form tissue
E.g.: Female bacteria reorganized by male bacteria
- Paramecia – different matins reorganize each other
- PM- glycoprotein act as Antigen(provide basis of immune response)
- Lipids – amphipathic in nature



- Lipids has kinks in their Fat tails
- Prevent close packing & make more fluid
- HO-O=O -Double bond causes kink in tail
- Protein of P.M



Act as receptor Act as enzyme

D. Function of Plasma Membrane

- Selectively permeable (Regulate the flow of materials into and outside the cell)
- Allow useful O₂ (or) Co₂ and useless & harmful products passed out
- Beet root does not lose color (P.M is impermeable to movement of Pigments molecule out of cell)
- Gases o₂ & co₂ diffusion through P.M
- Ions & small Polar molecule water allowed through P.M
- Water allowed through P.M by channel protein
- P.M active transport substances into cell affected by Cold Cyanide & absence of O₂, P.M contain carrier Or transport protein, Permease, Transporters
- Size of molecule can pass through PM 1- 15 Å
- Bulk transport of materials of PM by Exocytose or Endocytosis
- High Molecular weight (protein, polynucleotide & Polysaccharides)
- Transformed in to cell throw PM by Carrier protein.
- Ions- impermeable
- Water- freely Permeable
 - Endocytosis- in folding or extension of PM to form valuable or vesicle
 - Phagocytosis (cell eating) First observed by (Metchnikoff)
 - Pintocytosis (cell drinking) (taken up liquid)
 - Micropinocytosis (Micro pinocytic visible)

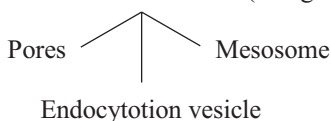
Receptor mediated Endocytosis
(Coated pit Endocytosis)

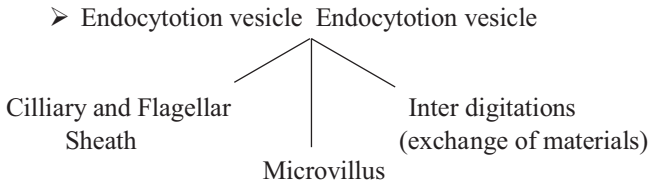
Specific protein + Substance- coated pit- investigate pintocytosis- form vehicle
recetosome

- Exocytosis : Reverse of Endocytosis
- : Reserve Pintocytosis

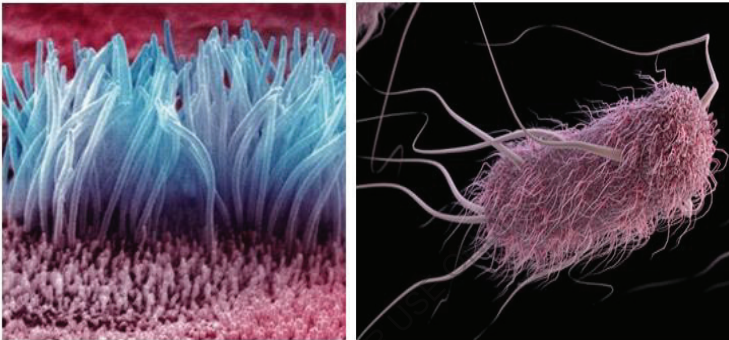
E. Modification of Cell Membrane:

- Folding- cell membrane infold (invagination)





4.3. CILIA & FLAGELLA:



- Cilia and Flagella are complex filamentous cytoplasmic structures protruding through a cell wall.
- They are minute, especially differentiated appendices of the cell.
- Flagella (singular = flagellum) are long, hair-like structures that extend from the plasma membrane and are used to move an entire cell.
- Cilia (singular = cilium) are short, hair-like structures that are used to move entire cells (such as paramecia) or substances along the outer surface of the cell (for example, the cilia of cells lining the Fallopian tubes that move the ovum toward the uterus, or cilia lining the cells of the respiratory tract that trap particulate matter and move it toward the nostrils).
- The terms cilium (meaning an eyelash) and flagellum (meaning a whip) are often used arbitrarily.
- Generally, cilia are shorter than flagella ($<10\ \mu\text{m}$ compared to $>40\ \mu\text{m}$).
- Cilia are present on the surface of the cell in much greater numbers (ciliated cells often have hundreds of cilia but flagellated cells usually have a single flagellum).
- The real difference, however, lies in the nature of their movement. Cilia row like oars. The movement is biphasic, consisting of an effective stroke in which the cilium is held rigid and bends only at its base and a recovery stroke in which the bend formed at the base passes out to the tip.

- Flagella wriggle like eels. They generate waves that pass along their length, usually from base to tip at constant amplitude.
- Thus the movement of water by a flagellum is parallel to its axis while a cilium moves water perpendicular to its axis and, hence, perpendicular to the surface of the cell.

4.4. MICROVILLUS:

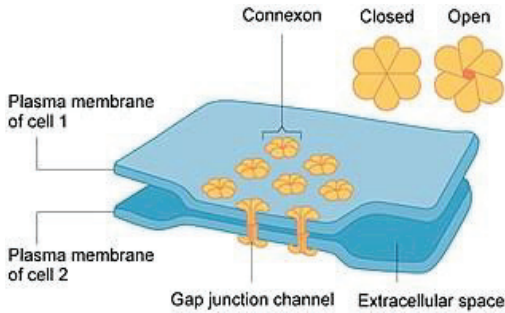
- 2000 Microvillus/ cell
- 2/μm long
- 800-1000Å⁰ thick
- Made up of protein Villin & fimbrin & Calmodulin
- It increases absorption surface seen Electron Microscopic Seen in light microscope striated brush border.
- Intercellular Junction Narrow gap between adjacent cells in animal tissue
- Intercellular space contains fluid
Desmosomes, Intercellular bridges, Terminal bars
(Minute projection arises from adjacent cell).

4.5 TIGHT JUNCTION

- Similar to Desmosomes the membrane adjacent – but lack tonofibrils
- Help conduction of electric signal in the animal cell held close.
- Electric current were to pass through the together (sealing strands)
- The material ties the cells & checks movement of materials
- Tight junction also cells Zonular occludens circular patches of cell
- Membrane held together inner side dense plate of protein of mechanical support

4.6. TONOFIBRILS:

- Radiating into the cell.
- Act as spot levels.
- It occurs between epithelial cells.
- It is called maculae adherences.
- Maintain stress tolerance of epithelial level.

4.7. GAP JUNCTION:**Figure 2: GAP JUNCTION**

Direct cytoplasm to cytoplasm connection between animal cells

- Gap junction filled with protein tube called connexions.
- Each connexions surrendered by 6 identical protein submit
- Permit ions & small molecule without leak in.
- Heart Muscle contain many Gap Junction – allow electric current heart muscle simultaneously.

4.8. PLASMODESMATA

- Inter cellular junction in plant tissues
- Cell wall & middle lamella have tubular passes
- Waste of small solutes may pass from cell to cell
- Certain protein & RNA molecule can also pass via Plasmodesmata.

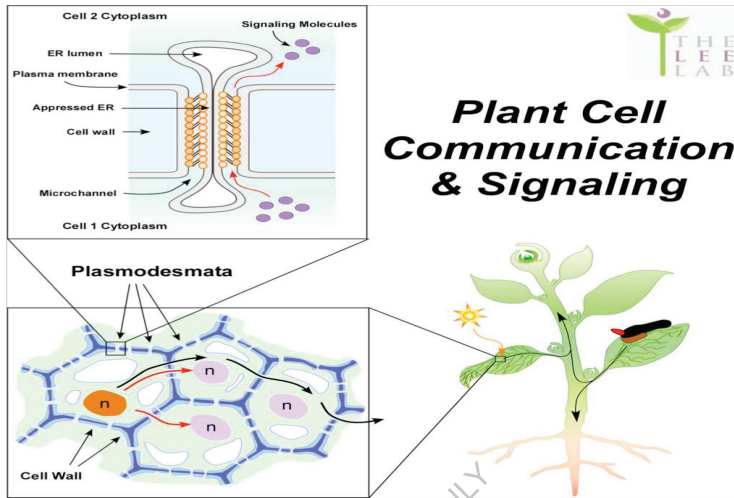


Figure 3: PLASMODESMATA

4.9 EXTRACELLULAR COATS

A. Chitin-Exoskeleton

- Hetero polysaccharides
- Protect the cell from injury & loss of water

B. Lycocalyx

- It covers epithelial lines of intestine
- It consists of Polysaccharides
- It protects cell from digestive enzymes.

C. Basement Membrane

- Compact sheet under epithelial cells
- Consists of collagen fibrils associated with glycoprotein
- Anchors cells to the connective tissue
- Acts as a selective diffusion barrier.

D. Cell Wall

- Rigid coat around plant cell
- Composed Polysaccharides
- Protect & support the cell
- Prokaryotic cells & Tonal Hyphase -cell wall made-up of Peptidoglycons.
- Cell wall made up of chitin.
- Cell wall-Slimy Polysaccharides- Polypeptide capsule

4.10. MICROTUBULES:

- Long, un branched , straight& rigid, Cylindrical element
- 25nm Diameter
- Made up of Tubulin
- Present only Eukaryotes
- Components of cilia & Flagella
- Spindle fibers duress cell division
- First observed by- De Robertis & Franchin(1953)
- Structure studied by- Led better & porter (1964)
- Central core(15 nm diameter)
- Longitudinal filament made up of tubulin.
- Microtubules easily broken down & easily reassembled.
- Microtubules maintenance cell from movement of chromosome movement cell Intermediate filament thickened between Microtubules & Microfilaments
- Made up of Keratin
- No of inactive metabolic by products cell inclusion called deutroplast or Metaplast.

4.11. CYTOPLASM:

- Cytoplasm term proposed by Rudolf Von Colliker
- Most of the cell activities occur in cytoplasm
- Cytoplasm homogeneous continuous phase called hyaloplasm= round plasma Cytoplasmic matrix is called cytosol.
- Cytoplasm-Ectoplasm or Plasma Gel (thin, moist rigid, Hyaline, Homogenous Portion.
- Cytoplasmic matrix contain the directional network- fine thread microfilaments(lattice filaments)
- Microtrabecular lattice- Keep cell organelles in position- acts as cytoskeleton

- Microfilaments or active filaments
- Cons, thin, Cylindrical rods 6-10 nm
- Responsible for cell mobility & Endocytosis
- Movement of microvilli
- Contraction of muscle fibers (in animal).
- Cytoplasm contains Nucleotides, Vitamins, t RNA, Enzymes many Minerals sugar amino acids.
- Cytoplasm plays intracellular distribution of Nutrients Enzymes
- Cytoplasm explained by - Strasbower 1882
- Cytoplasm matrix /Cytosol / itytoplasm
- Occur cyclosis or Cytoplasmic Streaming
- Cell-inclusion Starch grains concentric rings around Proteinaceous helms stored Rhizomes, Potato, rice, Maize glycogen granular liver, Muscle near SER
- Pigments granules or Chromatophores--Lipophores(Red, Yellow, Orange)
- Melanophores (Brown, Black)
- Fat droplets- present animal cell of adipocytes.

FUNCTION OF CYTOPLASM:

- Exchange of materials between cell organelles
- Site for Glycolysis FA Synthesis
- Intracellular distribution.

4.12. VACUOLE

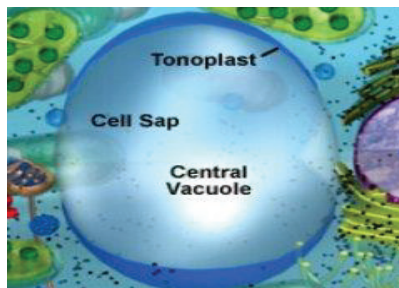


Figure 4: VACUOLE

- Term vacuole by Spallanzani.
- Covered semi permeable membrane called tonoplast
- Arise from ER/ Golgi complex
- Animal cell –Large no. of Vacuole
- Plant cell- large sized few vacuole, Absent in bacteria & PPLO
- Cell Sap contain water minerals gases sugar, organic acid wastes
Pigments(Autocyanin, Anthoxanthine)
- Vacuole act as Sap Vacuole ,Contractive Vacuole, Food Vacuole,
- Keep cells turgid of the cell.
- Vacuoles sap & contractive vacuole maintain Osmoregulation.
- Vacuoles contain Pigments help attract Birds & insect for Pollination
- Food Vacuole store minerals Carbohydrates
- Pigments Vacuole-Color to Flower root & stem of the plant.
- Store waste products & plant metabolites.

4.13. **ENDOPLASMIC RETICULUM**

- ER- Skeletal system of the cell.
- Network of 60 nm diameter
- Cris-Crossing like structure in the Cytoplasm.
- Present all eukaryotic cells except RBC in man
- Absent all Prokaryotes cells.
- ER-inner communicates system
- Cisternal, Vesicle, Narrow, theard, filled ,tubules
- ER----1945 discovered by Porter, Claude, Fullman, named by Porter
- Wall of ER single unit membrane
- ER fluid Endoplasmic matrix
- RER equivalent to Erastoplasm –arnier 1887.
- SER-continuous plasma membrane and membrane of Golgi cistern
- ER Fragmented small Vesicle called Microsomes.

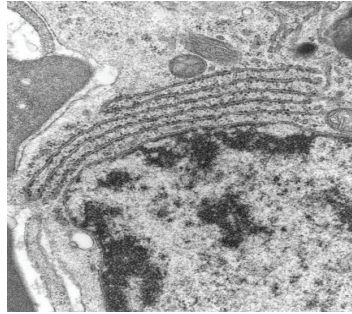


Figure 5: MICROSCOPIC VIEW OF ENDOPLASMIC RETICULAM

- provide mechanical support to the cytoplasm
 - Act as intracellular transport system
 - Accessory to the cytoskeleton
- RER- synthesize and secrete protein
- Specialized extensive membrane synthesis, e.g. immature cell (or) retinal rod cells.
- Oligosaccharide composed of N-acetylglucosamine, Mannose, Glucose.
- SER Specializes in lipid metabolism.
- Synthesize steroid hormones help of cholesterol e.g. (Testosterone cells)
 - Muscle cells contain SER called Sarcoplasmic reticulum

↓
Sequester Ca^{++} from cytosol
↓
Permit relaxation of Myofibril
↓
Followed Muscle contraction

- Main function ER-produce protein & lipids which help Synthesis of new cellular membrane.
- ER-called membrane factory (synthesized Protein & lipid and export other parts of cells.
- Major Phospholipids Synthesized in ER lecithin (Phosphatidylcholine)
- Transitional ER
- Ribosome free region of the rough ER are called transitional ER.
- Transitional ER from RER (instead of SER).

SER }
RER } Inside the liver act as detoxification of antigen

SER&RER

- Vesicle form Nuclear envelope at telophase stage of cell division
- ER provides space for Storage Synthetic Produce like glycogen.

Enzymes of Endoplasmic Reticulum

- Sterases, NADH cytochrome D, Reductase
- Mg^{++} activated ATP ase
- Nucleotide diphosphatase help to Biosynthesis of Phospholipids, Ascorbic acid, Glucuronide Steroids, hexose metabolism.

Function of Endoplasmic Reticulum

- Synthesis of glycerol
- Metabolism of Plasmolgens
- Synthesis of FA
- Biosynthesis of Steroids
- L-ascorbic acid Synthesis
- UDP-Uronic acid metabolism
- UDP-Glucose de Phosphorylation
- Aryl and steroid metabolism
- ER stained by basic stains.
- Named Erastoplasm, Basophilic in nature.
- Chromophilic substance or Nissl's bodies are present.
- Annulate Lamellate (pored ER)
- ER of invertebrate has pores
- ER of oocytes
- ER- of Spermatocytes of Vertebrate
- ER-Erastoplasm = ER (Garnier 1897)
- Endoplasmic reticulum was coined by Porter.
- ER- Act as a cell circulatory system, Cells vascular system.
- ER found Eukaryotic cells, Skeletal muscle (Sarcoplasmic)
- Newton called Nissl's granules
- Retinal Pigment Myeloid bodies.
- ER- absent in mature RBC-E-embryonic cells
- Evagination Nuclear membrane to form RER-SER.

Parts of Endoplasmic Reticulum

1. Cisternae

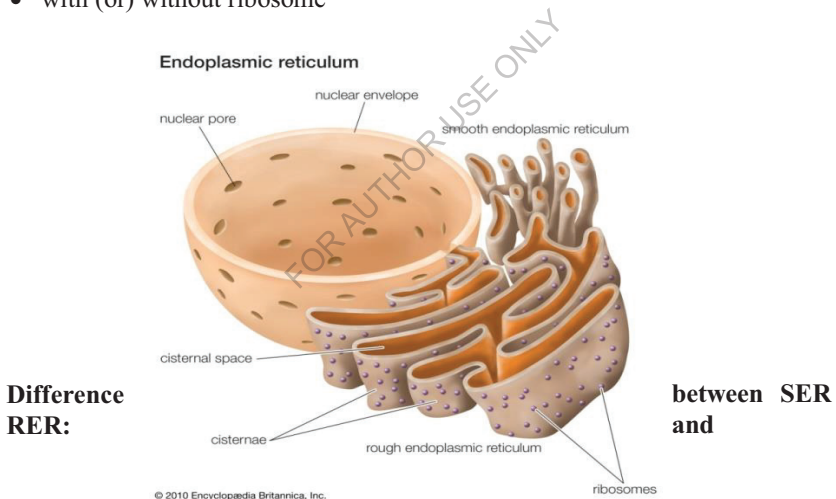
- Sac like structure.
- Un branched
- Parallel tubules
- Having ribosome

2. Tubules- Tube Like

- Irregular
- Branched
- Without ribosome

3. Vesicle

- Round(or) spherical
- with (or) without ribosome

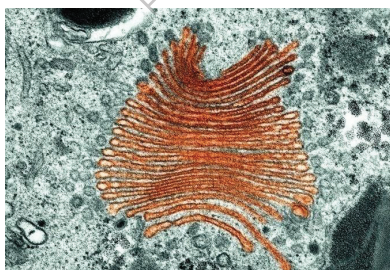


SER(Smooth Endoplasmic Reticulum)	RER (Rough Endoplasmic Reticulum)
• Devoid ribosome	• Have ribosome • 60s unit of ribosome • Attached with ER by ribophorin I & Ribophorin II
• More no. of tubules& Vesicle more	• No. Of Cisternae & tubules are more

• More near to P.M & Golgi complex	• Abundant near the nuclear membrane
• Rich in lipid & Sterol Secreting cells • E.g.: Adipose tissue intestinal cells • Muscles • Liver etc., Synthesis carbohydrate & vitamin C chose by detoxification of pollutant & carcinogen	

Functions of Endoplasmic Reticulum:

- Provide mechanical support
- RER → SER → Golgi bodies → Primary Lysosome out of cell by Exosmosis
- Many enzyme of ER help metabolic activities.
- Exchange materials between cell & environment
- Help formation primary lysosome synthesis of nuclear envelope.

4.14. GOLGI COMPLEX:**Figure 6: STAINING VIEW OF GOLGI COMPLEX**

- 1898-Named Golgi complex by Camilo Golgi
- 1954-Named Dalton complex by Dalton
- 1951- named Mitochondria by Baker

- Golgi complex Idiosome
- Internal reticular approach
- Golgi complex present Eukaryotic cells except RBC & Prokaryotes
- Golgi complex surrounded by Zone of exclusion
- Golgi complex absent ribosome / mitochondria
- Golgi complex originate from ER
- Two form of Golgi complex
- Localized form
 - Dictyosomes e.g. Plant Cells
 - Golgi complex
- Located fixed position of near nuclear
E.g. Vertebrate cells
- Golgi complex have definite Polarity
- Distal Maturins
- Concave region facing towards P.M Transface
- Proximal Formins (or) Convex from ER & cytosol Vesicle Cisternae.
- Made up of Cisternae (sac like, un branched 3-12 parallel of concentric tubules)
- Tubules (Tube like branched)
- Vacuole (Saclike present near maturing face).

Chemical Composition of Golgi complex:

- Golgi membrane rich in Phospholipids
- Enzyme-AD pase, ATPase, CTpase, TPP- Thiamine Phosphatase, NADH-Cytosome C reductas, G-6 Phosphatase.
- Carotenoids, Fatty acid & Vitamin.
- Main enzyme of Golgi complex: Nucleoside Phosphatase.
- Golgi complex otherwise called - Lipochondrian, Dalton complex, Idiosome, baker's body.
- Golgi complex absent in prokaryotic cell
- Golgi complex presents all eukaryotic cells.
- Golgi complex absent mammalian RBC, sperm cells of bryophytes.pteridophytes, sieve tube of plants-absent Golgi complex.
- Golgi complex function - secretory, Absorptive cells.
- Nerve cell Golgi complex surrounds the Nucleus.
- Golgi complex - Cisternae, Golgi vesicle, cisternae arranged like stack of plate called a Golgi stack (or) Dictyosomes (or) Golgi some.
- Cisternea-functional unit of Golgi complex

- Plant cell= Golgi stacks or Dictyosomes
 - Golgi complex have polarized structure
 - Convex - cisternae face or cis - face
 - Vesicle derived from smooth ER formed from Cisternae of Golgi complex
 - Concave face of Golgi complex called mature and trans - face towards the P.M
 - Maturing face of Golgi complex breaking off into vesicle called secretory Vesicle
 - Golgi complex
 - ER
- } End membrane system term was coined by De-Robertis.
- Nuclear envelope formation.
 - Golgi complex with help of secretory vesicle, transporting secretory substance
 - Golgi complex - Principle director of macromolecular traffic in cell.
 - Main function of Golgi complex- secretion and transport
 - Golgi complex → Secretory Protein → Secretory vesicle
 - Protein Zymogen granule (Exocytosis Peroxisome)
 - Golgi complex modified macromolecules by Oligosaccharides
 - Glycosylation of lipids - Glycolipids
 - Glycosylation protein carbohydrates- glycoprotein
 - Sulphation Glycoprotein- (Cortical cells)
 - Golgi complex contain enzyme glycosyl transferases & manosidases
 - Golgi vesicle Secrete carbohydrates for synthesis of cell wall or cell plate
 - Important Glycoprotein Secreted by Golgi apparatus Mucin Form Mucus
 - Secreted by goblet cells of respiratory and intestinal Epithelium
 - Root cap cells of plants Contain Golgi bodies secrete mucous Polysaccharides acts as lubricants
 - It helps to penetrate the soil.
 - ER-Synthesis Lipids -transport through Golgi complex
 - Golgi complex important role formation of Lysosome
 - Acrosome of sperms
 - Formation granules.
 - Secretion hormone insulin, tyrosine.

Functions of Golgi complex:

- Intra cellular network
- Modifies like protein carbosterolect
- Form secreting vesicle discharge their contents to exterior. E.g. Tears, Sweat, Saliva Hormone.
- Carryout membrane formalin.
- Formation of Lysosome.
- Formation of Acrosome in sperms.
- Carryout recycling of P.M.
- Production of hormone from endocrine glands (Thyroxin- from Thyroid gland)
- Produce pigment granules in Tumor or Cancer cells.
- Produce Lysosome or Suicidal bags Lysosome are disposal units. Discovered by Christian de Dave 1955. Membrane bound vesicle contain Hydrolytic enzyme
- Lysosome Form of minutes crystalline or semi crystalline granules. It involved intracellular digestion e.g: WBC in blood.
- Lysosome involved Phagocytes in liver.
- Histolytic in connective tissue osteoclast in bones.
- Lysosome contain 40- 50 types of hydrolytic enzyme(Synthesized from RER)
Eg:
 1. Nucleaser- Diestins N.A (Nucleic Acid)
 2. Lipases- Diestins lipid
 3. Phosphatase- Phosphate linked compound
 4. Sulphatases-Sulphur linked compound
 5. lycosidases- Polysaccharides digestion etc..

4.15. LYSOSOME:

- Lysosome (Lysis=splittins+ soma =body)
- Abundant in animal cells (Phaoustil activity)
- Lysosome absent in RBC cells.
- Plant cell large central vacuole act as Lysosome.
- Lysosome found in roof tip cells of Maize (Particularly in dyins cells)
- Lysosome found yeast & fungi – absent in Prokaryic cells
- Lysosome first discovered by Christiande Dure. (Belian Cytocovered)1955.

Basis of Biochemical Studies of Lysosome

- 1956-Lysosome observed with electron microscope by Novikoff.
- Lysosome –Size –Single membrane bound organelle..
- Lysosome buddle off from Golgi cisternae or ER Tubules.
- Lysosome reservoir of hydrolytic enzyme.
- Lysosome involved intra cellular digestion.
- 40 enzyme in Lysosome....(Protease, Nuclease, Glycosidase, Phosphatase, Lipases, Phospholipase)
- Acid Phosphatase important enzymes in Lysosome.
- Acid Phosphatase enzyme to maintain PH 5.0 within the Lysosome.
- RER- transported to Golgi complex packed Lysosome.
- Golgi Vesicle (processed enzyme)----Vesicle bud off called primary Lysosome or (protosomes) (Contain only one types of enzyme)---Endocytosis (Digestion of materials)--Fuse with vesicle (or)vacuole Secondary Lysosome (otherwise called digestive vacuole & Food vacuole----Digested materials absorbed by cytosol -- Secondary lysosome converted multivehicle body-- Excreted by exocytosis.
- Autophagy-Digestion surplus organelles.
- Primary Lysosome + Autophagy vacuole
↓
Autophagy
Eg: Metamorphosis in insects.
- Metamorphosis in amphibian.
- Germination of seeds during starvation.
- Release enzyme outside cell-Exocytosis
E.g.: Remodeling of Cartilage bone
- Remodeling of bone (Osteoclast)
- Autolysis(self destruction)
- Lysosome are called suicidal bag
Eg: Resorption of tadpole tail during Metamorphosis.

Lysosome arises from Golgi complex or SER

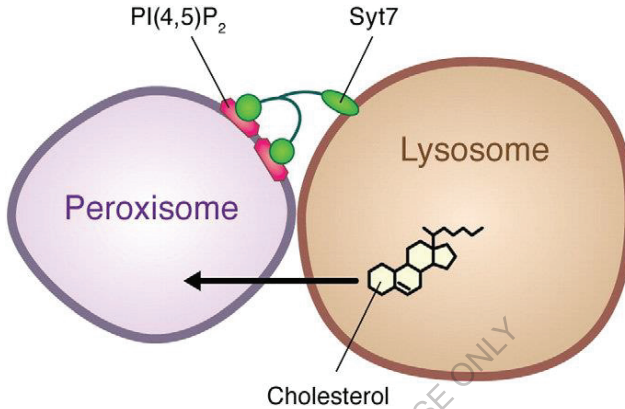


Figure 7: LYSOSOME AND PEROXYSOME

Lysosome is Polymorphism in nature there are...

1. Primary lysosome (or) Storage granules (Vesicle filled with inactive hydrolytic enzyme).
2. Secondary lysosome (or) Heterophagosome (lysosome fused with ingested food).
3. Residual body / Tertiary lysosome (lysosome only with undigested food)- thrown out exocytose.
4. Auto lysosome (or) Auto phagosome – Primary lysosome feeding cell organelles (or) worn-out cells.

Types of Lysosome

- Primary Lysosome
- Secondary Lysosome
- Residual bodies
- Auto phagic vacuole
- Absence of Lysosomal hydrolase occur serious hydrolytic enzyme causes- Hurler s disease.

- Bones deformed by accumulation glucosamino glycol
- Lysosome Stabilisers
- substance which strength the Lysosome membrane
Eg: Cortisone, Cortisol, Cholesterol and Histamine, heparin.
- Lysosome labellers (substance weaken the lysosome)
Eg: Progesterone, Testosterone.

Enzymes of Lysosome

40 Different Enzyme or hydrolyses

1. β -Glycuronidase
2. β -galactosidase
3. α -Glycosidase
4. β -N- acetyl Glycosaminidase
5. α -Mannosidase
6. Cathepsin-A and Cathepsin $-\beta$
7. 7Deoxyribonuclease
8. Acid ribonuclease
9. Acid Phosphatase
10. hyaluronidase
11. Phospholipase
12. Phosphoprotein

Functions of lysosome:

- Meant for intracellular digestion like break down of worn out cells, organelles & components
- Digest extracellular foods
E.g. Acrosome produce hyaluronidase to digest egg envelope, osteoclast cells digest bones etc.
- Digest harmful substances (defense of the body).
- (Phagocytosis) e.g. Neutrophils engulf foreign particles.
- Involved release of hormone e.g. Thyroxine from thyroglobulin.
- During metamorphosis digest unwanted organelles
E.g. 1. Tail & Gills in tadpole
2. Enlarged uterus after parturition.
- During starvation - hydrolyze organic food - provide extra nourishment.

- Cause breakdown of nucleic acid and result in mutation.
- Residual bodies accumulate in the cells leading to gout, silicosis, rheumatoid, asbestosis, and irritation.
- It can reduce Aging. E.g. rotting of fruit reduced by checking action of polygalactouronase.

4.16. NUCLEUS- KARYON:

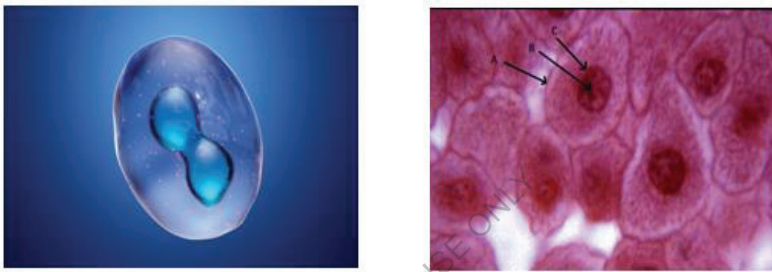


Figure 8: STAINED NUCLEUS

- Nucleus visible high power of Microscope.
- Fine structure observed with help of (Electron Microscope) Observed by Dutch Microscopist- Anton Von Leuwenhoek
- RBC cells of fishes- Scotch botanist (Robert brown) first described the Nucleus orchid cell 1831.
- RBC-no nucleus- survives few months.
- Phloem cells of sieve tubes – Function several year
- Nucleus control growth & heredity proved by experiment.
- Nucleus essential for survival of the cell.
- Uninucleate - single nucleus eg. Amoeba
- Bionucleate- Paramecium Caudatum.
- Multinucleate- animal & plant cell E.g.: striated muscle fiber latex vessel.

- Multinucleate cells formed Syncytia e.g. opalina Epidermis or nematode striped muscle
- Multinucleate cells formed Plasmodia or eg: Rhizobustursi Alae Vaucheria
- Position of Nucleus - gland cells (Nucleus at the base)- Mature Plant cells (one side from the centre) Adipose cells.

Shape of the Nucleus

- Flattened Nucleus – Squamous epithelial cells.
- Rounded- Cuboidal epithelial cells.
- Oval-Columnar epithelial cells.
- Bean shaped – Paramoecium.
- Horse Shoe Shaped- Vorticella.
- Elongated – Smooth Muscle Fiber.
- Monilliform-Stentor.
- Branched- Silk Spinning cell of insect larva.
- 2-lobed- Eosinophil.
- 3-lobed- Basophil.
- Many lobed – Neutrophil.
- Lens Shaped- Amoeba.
- Size of Nucleus 5-25 μm in diameter.
- Nucleus- Karyotheca (outer membrane) - Karyo Lymph-Nuclear matrix - one or more Nucleoprotein filament round body. e.g. chromatin Nucleoli
 - Nucleus composed -80% protein
 - 12% Dna-5%-RNA, 3% lipid
- Nucleus contain enzyme Polymerase (help in DNA synthesis) and (Ca, Na, K, M, Phosphorus, DNA & RNA).
- Nuclease-Cell maintenance & growth Cell Metabolism
- Functions of Nucleus (Genetic information, Cell replication, and Ribosome formation Variation and Differentiation cell.
- Nucleus envelope or Karyotheca consist of 2 unit membrane
- In-between two membrane Per nuclear space (150-300)-continuous channel to ER
- Nuclear envelope- Semi permeable-Minute pores (1000-10000 per/ Nucleus
- Size of Nuclear pore (300-1000 $\text{\AA}$)

- Each Nucleus pore fitted with protein structure called Pore complex
- Pore complex control passage of macromolecule such as +RNA, mRNA, ribosome protein Nucleotide Enzyme ribosome submits.
- RNA (hydrophilic) only passed nuclear envelope
- Nuclear sap otherwise called Nucloplasm
- Nucloplasm term by Karyolymph
- Nucloplasm contain raw material of –Nucleotide , Enzyme, metal ions(Mn^{++} , Mg^{++})
- RNA and Ribosomal Subunits Synthesized in Nucloplasm.

FUNCTION OF NUCLEUS

- Nucloplasm seats of synthesis of DNA and RNA chromatin materials and nucleoli.
- Provide turgid to the Nucleus.
- Nucleus matrix is a network of fine Protein Containing fibrils. Occur DNA replication transcription maintain shape of the nucleus.
- Chromatin fiber anchored with Chromatin.
- Chromatin means colored nuclear materials.
- Chromatin during inter phase stage called as Chromatin fiber.
- The term chromatic network given by Fleming- Chromatin reticulum
- Chromosome centromere appears as a narrow vein called Primary Constitution.- Secondary constriction-beyond secondary constriction-called satellite-SAT Chromosome or marker Chromosome.
- Secondary Constriction site for formation of Nucleoli- Nucleolar Organizers.
- Nucleolus otherwise called little Nucleus or Ribosome Factory of the cells
- Chromatin two types – Euchromatin (light stained)
- It permit Crossing over –posses control part of Nucleus
- It replicate early S phase of cell division.
- Heterochromatin dark stained body more histone & less acidic, genetically less active
- Stop point in transcription.
- It replicate late S phase.
- Posses near Nuclear membrane.
- It inhibits crossing over.

4.17. CHROMOSOME

- Chromosome not visible interphase stage (Chromatin reticulum)

- Metaphase stage Chromatin materials condensed form chromosome.
- Chromosome first observed by Hofmeister 1818 and Karl Naeli [Pollen Mother cell of *Trandes cantia*]
- Chromosomes Structure described by Strasburer.
- 1875-credit of discover of Chromosomes by Strasburger.
- Term Chromosomes proposed by Waldeyer 1889
- Term Choromotin suggested by Flemming
- Chromosomes best studied at Metaphase Chromatin-Choromatin thread-Gelatian-dehydration- Coilins- Clear Chromosome.

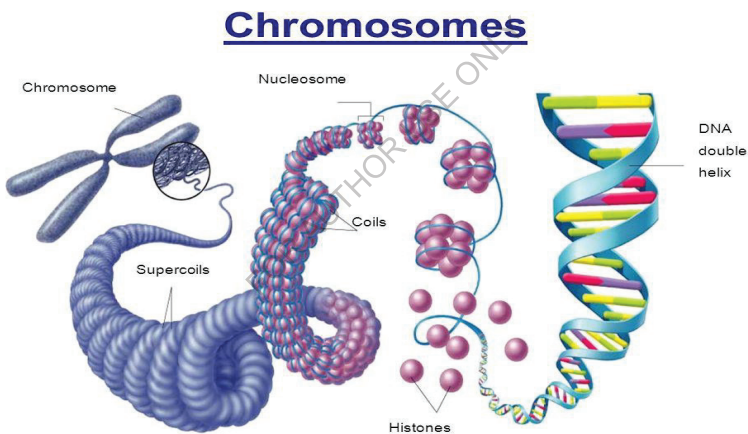


Figure 9: STRCTURE OF CHROMOSOME

Human cell contain $2n=46$ -reported by T.H Tijo and A. Levan (1950).

S.NO	PLANTS	2N	N
1.	Pisum sativum(Pea)	7	14
2.	Maize (Zea mays)	10	20
3.	Wheat(triticum)	21	42
4.	Ophioglossum reticulatum (Pteridophyta)	631	1262
5.	Ascaris(Round worm)	1	2
6.	Drosophila(Fruit fly)	4	8
7.	Chimpanzee	24	48
8.	Aulocantha (Protozoa)	800	1600

- A Chromatid contains single fine Chromatin fiber Chromonema.
- Chromosome contain unit of heredity gene
- Metacentric Chromosome – centre centromere and two equal arm
- Submetacentric Chromosome - centromere slightly away from the middle-one arm slightly short other arm slightly long.
- Acrocentric chromosome -Centromere is short other arm very long
- Telocentric Chromosome -Centromere at the tip of chromosome- centromere at the tip of Chromosome arm one side only.
- NOR or Nucleolus organizer or Secondary Constriction of Chromosome Present in 12, 14, 15, 21, 22 Chromosome in human.

4.18. ANEUPLOIDY IN MAN

(chromosome basis of position of centromere.

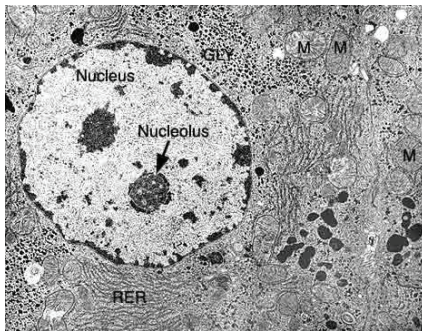
- 21-trisomy-Downs Syndrome.
- 18-trisomy- retard Children.
- 13-trisomy- Patau Syndrome.
- 21-Monosomy- opposite to Mongolism.
- 18-Monosomy- Deletion one Pair
- Klienfelter Syndrome-(44-XXY)
- Turner s Syndrome-(44 XXO)

4.19. NUCLEOSOME:

- Chromatin fiber Chain of similar subunits called Nucleosome.-consists of core particle wrapped by DNA stored.-core particle octamer of histone molecule.
- Nucleosome connected to next short DNA linker-linked together Chromosome.
- Chromatin forms Chromosome during cell division.

Difference between chromatin and chromosome:

CHROMATIN	CHROMOSOME
• Visible interphase Nucleus	• Visible in M phase(phase of Nucleus division)
• Un condensed form of deoxyribnucleoprotein chain	• Condensed form of deoxyribonucleic protein chain
• Filaments lying Crisscrosses forming Chromatin reticulum	• Appear short thick rod like organelles
• Control metabolism	• Control genetic information to the cell
• Replication of DNA occur in Chromatin phase	• Replication of DNA suspended in Chromosome phase

4.20. NUCLEOLUS:**FIGURE 10: NUCLEOLUS**

- Nucleolus discovered by Fontana-178.
- Nucleolus described by Waner 1840.
- Nucleolus names by Bowman 1840.
- Nucleolus dense rounded dark stained, granular organelle.
- Nucleolus consists of largely RNA & Protein contain DNA which produce r RNA
- Nucleoli formed Specific sites called Nucleolar organizer of Chromosome
- Calcium (Ca^{+}) essential for maintain configuration nucleolus.

Nucleolus four regions

1. Matrix-Homogenous, Proteins round substances
2. Febrilar Region Fine fibrils Present RNA Precursor.
3. Granular Region –contain ribosome submit rRNA + ribosome protein.
4. Nuclear Chromatin fiber it synthesis r RNA.

Function of Nucleolus:

1. Synthesis and store RNA
2. Receive ribosomal Protein the cytoplasm for store
3. Ribosome submit formed wrappins rRNA + ribosome Protein-Nucleolus through Nucleopore Cytoplasm
4. Play a role in cell division
5. Nucleus-it contains genetic information of vehicle of heredity.
 - Reproduction
 - Development
 - Metabolism
6. Nucleus- Direction for control cellular activities through formation mRNA
7. Nucleus take place Production of ribosome Nucleus Hannerlin to Prove control tower of all hereditary information.
8. Experiment on Acetabularia-Uninucleate green algae.

9. Nucleoid contains high concentration of Polyamines-Spermidine-cadaverin.
10. Acidophilic dense granule-Nucleolus
11. Nucleolus first described in Sperm cells-blastomere-muscle cell.
12. More synthetic activity large Nucleolus oocytes, Secretory cells.

Nucleolus Contains:

- 5-10% RNA
- rRNA and concentrate of thophosphate-Enzyme Acid Phosphatase, Nucleoside Phosphogles
- NAD⁺ synthesize enzyme-no DNA no Histone-RNA transmethylese present.
- Telomere of Chromosome provide Polarity of Chromosome.
- Centromere or Kinetochore is called primary constriction.
- More calcium ions in the nucleolus.
- Maintain intact organization of the Nucleolus.

Nucleolar Cycle:

1. Prophase stage – Nucleolus disappear
 2. Telophase stage- Reappear of cell organelle.
 3. Nucleoli cycle associated with chromosome Nucleolus organizer.
- Nucleolus – factory of biosynthesis of ribosome
 - Store all protein of ribosome and r RNA
 - 45s rRNA-18s-metylation –nucleolus
 - 28s r RNA (Ribosome protein & r RNA)
 - Karyoplasmic Index (KP) Nucloplasmic index(NP)
 - NP=volume of the Nucleus/ Volume of the Cytoplasm
 - Chemically Nucleus consists of 80% protein- 20% DNA, 5 %RNA, 3%lipids.
 - Mammalian cell nucleus contains 3000-4000 pores/ nucleus.

- Blebbin of Nuclear membrane described by Gay 1955.
- SAT Chromosomes – SAT (Sine Acid Thymonucleinic).

Chapter 5

Cancer Cell



Figure 11: CANCER CELL

- Cancer
- Latin word meaning- Crab
- Study of cancer called Oncology
- Abnormal tissue growth - neoplasm-can cause spreading- called as Malignant
- Types of cancer-Benin tumour (harmless)-Malignant tumour(harmful)
- 1775 Pott Cancer in Chimney Sweeps
- Shale oil-skin cancer in workers
- Radioactive ore-lung cancer
- Beta naphthylamine – Bladder cancer(rubber industry)
- Cigaretters- lung cancer
- Rouse sarcoma virus- sarcoma cancer
- Recently human T-cell leukemia- cause virus HTIV-1 (blood cancer)
- Normal- cancer cell (carcinogenesis).

5.1. CHARACTERISTIC OF CANCER:

- Tumor arises from existing tissue (or) cell of the body.
- Growth of tumor autonomous
- Tumor cells undifferentiated & anaplastic in nature.
- Anaplasia- reversion of differentiated cells into undifferentiated (or) embryonic cell.
- Greater Potentiality for growth and Multiplication
- Cancer cell not any specific function
- large and irregular nuclei
- Lose their contact inhibition.
- cancer cell Properties
- Cancer does not follow law of Proliferation (or) cell division.
- Cancer cells avoid apoptosis(programmed cell death)
- Cancer cells are immortal.
- Cancer cells have dedifferentiation (Cancer cells return to undifferentiated condition.
- Cancer cells Proliferate indefinitely.(uncontrolled Proliferate)
- Metastasis (Cancer cells spread to their tissue (or) organs.
- Mutation- Normal cell – Cancer cells mutagens like chemical (or) x-ray.
- Cancer cells – Genetically unstable
- Cancer cells lose their ability to repair damaged DNA.
- Contact inhibition (Normal cells stop Movement and division contact with each other.
- Cancer cells no contact inhibition
- Nucleus – Larger in size & irregular in shape
- Nucleolus_ Large in size exhibit Hypertrophy (enlargement of tissue (or) cell.
- Cancer cells – abnormal mitosis – Chromosome Eg: Myeloid leukemia Patients.
- (22th Chromosome loss arm of Lost)- Chromosome short (Philadelphia Chromosome).

5.2. GENETIC BASIS CANCER:

ONCOGENES

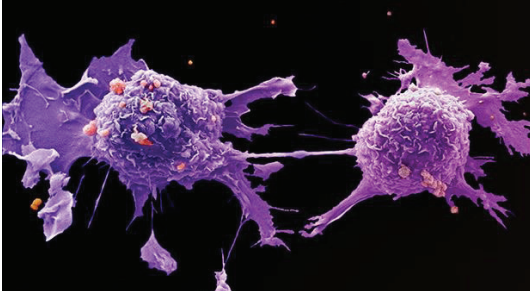
- “Oncos”—Greek word mass (or) tumor
- Cellular Oncogene is called proto Oncogene.
- Oncogene originally discovered in retro virus
- Oncogene in retro virus is called viral Oncogene -produce cancer growth.
- Viral Oncogene found -Adeno virus, Papura virus, Herpes viruses, and Retro viruses.
- Oncogene also found in all human cells is called cellular Oncogene.
- Cellular Oncogene activated by viral Oncogene -produce cancer growth.
- Oncogene are not found in RBC because not in nucleus
- 20 Oncogene have been discovered in viruses C-myc, N-myc, L-myc, C-abl, C-myc b, C-erb B, C-K ras
- Infect RNA virus ---- DNA produce copier ----integrated with host cells --- chromosome ---integrated DNA contain Oncogene -----produce cancer.
- Oncogene is activated in Human tumor.
- Normal cell growth controlled by Suppressor gene.

Proto-Oncogene

- Part of genome involved activated-DNA Malignant induce-Ontogeny (Malignant growth)-Point mutation on DNA- Cancer.

ONCOGENS OF HUMAN TUMOUR:

ONCOGENE	TYPES OF CANCER	ACTIVATION MECHANISM
• Hox-11	• Acute-T-cell Leukaemia	• Translocation
• Erb B-2	• Breast-Ovarian Carcinomas	• Amplification
• L-myc	• Lung-Carcinoma	• Amplification
• Ret	• Thyroid Carcinoma	• Rearrangement

5.3. TYPES OF ONCOGENE AND CANCER:**FIGURE 12: SCANNED CANCER CELLS**

GENE	TYPES OF CANCER
• Apc	• Colon/ rectum Carcinoma
• BRCA1	• Breast and Ovarian
• 1 NK 4	• Melanoma, Lung Carcinoma, Brain tumor, Luminas Lymphoma
• Rb	• Retinoblastoma
• PTEN	• Brain tumors, kidney and Lung Carcinomas

Example of Carcinogens

- Polycyclic hydrocarbon -tars- benzpyrone
- Mustard gas (CH_2 , $\text{CH}_2 \text{Cl}_2$).
- Nitrogen mustered ($\text{CH}_3 \text{n}(\text{CH}_2 \text{CH}_2 \text{Cl}_2)$)
- 4. B-Naphthylamine and aminophenol
- Ionizing radiation from x-rays certain radioactive substances..

5.4. Types of Cancer in Organ Level.

- Sarcoma- malignancy in Structural tissues (connective tissue),
Eg: Osteosarcoma(bones), Liposarcoma(fat)
- Carcinoma- Epithelial Cancers, e.g.lung Carcinoma, Breast Carcinoma

- Lymphoma- Lymphatic tissue
- Leukemia- Proliferation WBC (Blood Cancer).

Etiology (Causes of Cancer):

No. of Chemical substances called Mutagens cancer called Carcinogenic agents.

- Microscopic observation (tissue biopsy).
- Cytochemical Methods (study of markers on the surface of the cells)
- Cytogenetic changes.
- Scanning and X-rays diagnostic methods.

5.4. NATURE OF CANCER CELLS:

A. Centrioles of the Cancer Cell

- Mis behavior of centrioles in cancer cell.
- Two centriole are separated by long random distance
- Do not have any spindle fibers.

B. Plasma Membrane of the Cancer Cell

- Most visible property of Cancerous cells invasiveness of other tissue
- It become leaky
- Tumors cell contain tumor associated antigens
- Cancer cell gap junction are disappear
- Cancer cell Plasma membrane lost Fibronectin (Glycoprotein)
- lost contact to another cell.
- Cancer cell release excessive anonomous of growth promotins factor – leakage cell membrane- dramatic Spurt in growth.-cell Multiplication of normal cell- become abnormal shape.

C. Cell Organelles in Cancer Cell:

- Ribosome & RER most abundant Cancer cell
- Golgi complex & Lysosome absent.

D. Metabolism of Cancer Cell:

- Glycolysis – faster rare

- Cancer cell use more blood sugar
- Fourth (or) later Mutation - cells are obviously deranged. (Malignant cells)
- Tumor Progression
- Man Cancer- Skin, Lungs, Prostate gland , Stomach, Intestine, Mouth
- Woman- breast, Uterus, intestine, Stomach& skin.

5.5. STAGES OF CARCINOGENESIS PROCESS:

1. Initiation
2. Growth
3. Promotion
4. Conservation
5. Propagation
6. Progression.-invasion-metastasis.

1. Initiation

- Initiation by-Carcinogens'-Viruses Heredity- end Mutation.
- Normal cell converted into Cancer cell –By Transformation, Mutation-Chromosome aberration, Mutagens-(Chemical, X-rays Viruses).

2. Promotion

- Cancer cell---Progressive growth occur--- cell stimulate to divide (disappear laws of cell division)---mass of cells (clone)---similar cells inherited in the Mutation of original cells.
- Tumor to developed by cannot stop cell division- do not undergoes differentiation.

3. Progression

- Final stage development of Cancer
- Cancer cell loss contact inhibition
- Leak out and enter blood Vessel- Carried by – blood- Deposited new place- Proliferates produce Cancer.
- New Cancerous Growth cells from one place to new place-Metastasis.

5.6. DURING PROGRESSION 3 IMPORTANT EVENTS

1. Loss of Contact Inhibition

- Dividing cell stop dividing
- Contact with other cells---- Loss of contact ---- do not stop cell division because Cancer cell loss their contact inhibition.

2. Metastasis

- Spread of tumor cells – established their secondary growth (metastasis).
- Cancer cell---- Secrete protease enzyme called Plasminogen activator--- activate serum protein Plasminogen--- then converted---- Plasminogen-Digest basal lamina of epithelial & endothelial eroded basal lamina--- Cancer cell move freely in blood stream---Entered blood in the lung, kidney etc.
- Start secondary growth-Lymph node.

3. Angiogenesis

- Angiogenesis means formation of new blood vessel on the cancer cell.
- Primary and Secondary tumor reuse blood vessel-need for nutrient & O₂.

5.7. THEORY AND HYPOTHESIS OF CANCER:

- **CONNIE'S THEORY** of embryonic cell rest early development of the embryo is left of cells may be misplaced- become intermingled with other tissue- accuse power of proliferation and growth.
- **HYPOTHESIS OF EALDEYER AND THIERSCH.** Normal condition static Equilibrium between epithelial tissue and tissue-Proliferation capacity is Checking the other due to age this lose of checking capacity- Proliferate rapidly Carcinoma.
- **HYPOTHESIS OF ANOPLASIA-** Hansmann Differentiated cells – embryonic type Or anaplasia- Production of more Primitive type-greater disease of anaplasia-greater malignancy.

CELL & MOLECULAR GENETICS

- **THEORY OF CELL VARIATION** - Haddow Carcinogenic agent inhibit normal cell function- injured cells live rise to new race of cells with increased rate of fission.-carcinogens act as enzyme poison- - produce new enzyme- Capacity for unlimited growth.
- **IMMUNOLOGICAL THEORY**: carcinogen with special cellular component- to form antigenic complexes-leading Immunological reactions destroys cells those cells.-it favor the emigrants and Proliferation of cells.
- **DEFECTIVE KREBS CYCLE**: Warbur 1956 cancer causing owing cellular respiratory apparatus- affect aerobic respiration- increased aerobic respiration-low energy released- incapable of maintains organization of cell- cancer growth uncontrolled.

5.8. TYPES OF CANCER

1. Malignant tumor
 - Grow rapidly
 - Spread other parts of the body
2. Non- Malignant tumor (Or) Benin tumor
 - Confined to the area
 - Its growth is always slow.

5.9. Types of cancer based on affected part of body:

TYPES OF CANCER	AFFECTED PART OF THE BODY
• Cacinoma	• Epithelial cells of skin & organs
• Sarcoma	• Malignant tumor in connective tissue
• Osteoma	• Tumor of bones
• Fibroma	• Tumor arises from fibrous tissues

• Glaucoma	• Network of supporting connective tissue in brain & CNS
• Melanoma	• Growing Pigmented lymphoid organs, eg: lymph node
• Lymphoma	• Cancerous growth lymphoid organs eg: lymph node
• Teratoma (Parasitic Foetus)	• Teratomas arise in cell- rest derived in morula stage of suppressed germ cell Fertilized Polar body it develop by Parthenogenesis

5.10. CAUSES OF CANCER**1. Chronic Irritation**

- Irritation caused to the tongue by broken (or) malformed teeth
- Cancer of the lip- Clay Pipes were in use.

2. Atmospheric Pollution

- Metropolitan cities- combustion of coal & oil with motor exhaust gases.-increase Cancer cases.

3. Smoking

- Heavy Smoking of Cigarettes –more no of lung cancer cases.

4. Radiation

- Excessive x-ray, cosmic rays causes Cancer.

5. Chemicals:

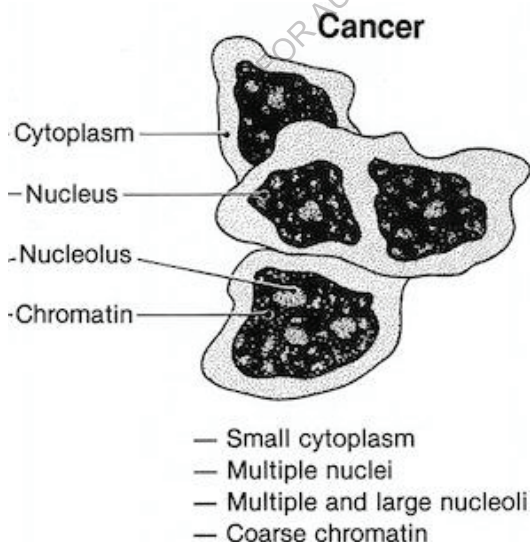
- Food, Medicine & flavours.

6. Diagnosis of Cancer

- X ray used to diagnose cancer digestive tract, kidney & chest
- Biopsy- suspected cancer tissue.

5.11. TREATMENT OF CANCER

1. Surgery.
2. Radiation –X-ray (or) radioactive substance
3. Laser treatment – Reddish black drug called Haematoporphyrin derivative (HPD) -
-----retained only by cancer cell -----1-3 days-----loaded with drug.
4. Dr .Alan Wilie said your tumor cannot detect x ray. Traced with blue laser light.
5. Cancer attacked by focusing a red laser beam ---- Activate HPD ----To kill malignant cells (Mr. Berne).
6. Magic Bullet Magic Bullet used as monoclonal antibodies.
7. Tumor cells have tumor antigen on their surface -----Ab prepared invitro-----Ab are toxin kill the cancer cell. Eg: “ricin “is a toxin of β -lymphoma cell.
8. Antibody linked to the ricin function called Magic Bullet---triggered in to blood---
--Combine with antigen of tumor cell ---- toxin diffused with cancer cell ----
killed by the toxin.
9. Surgery, Chemotherapy, Radiotherapy, Hormonal treatment was needed.

**Figure 13: DIAGRAMATIC CANCER CELL**

Chapter 6

Cytoskeleton Structure

- Cytoskeleton structure observed EM by Koltzoff 1928.
- Microtubules are found axon of modulated nerve fibers first seen 1953 by Robert & Franchi, he called as Neuro tubules.
- Microtubules found all eukaryotic cells of cytoplasmic matrix

6.1. MICROTUBULES:

Microtubules Present in Cell Organelles, Cilia, Flagella, Centrioles, Basal bodies, Mitotic apparatus, Sperm tail, Process of nerve cells supporting elements of protozoan.

- Microtubules absent in amoeba, Slime moulds
- Prokaryotic cell absent in microtubules
- Size of microtubules 25nm wide, 0.2-2.5µm long .
- Microtubules made up of protofilaments ----made up of globular submit tubulin ----one α tubulin and one β tubulin.

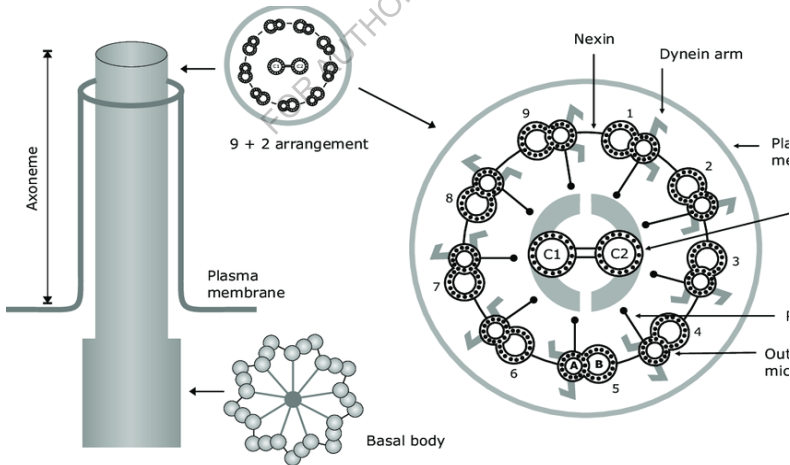


Figure 14: ULTRA STRUCTURE OF CENTRIOLE

.Origin of Microtubule:

- Tubulin subunit abundant in cytosol
- Increase tubulin subunit during mitosis & meiosis
- Colchicines----inhibit assemblage of microtubules ---prevent formation of spindle fibers.

Functions of Microtubules:

- To maintain shape of the cell
 - Motile elements like cilia and flagella
 - Help formation of spindle fiber
 - Help chromosome movements (or) chromatid in anaphase
 - Cause movements of chromosome at anaphase
 - Cause movements of pigment granules in the chromatophores
 - It components of centriole, basal bodies.
 - Intracellular transports
 - Serve as track of transported macromolecules & organelles
 - Lysosome
 - Mitochondria
 - Vesicles
- } from ER & Golgi complex
- Micro circulatory system (active transport use ATP).
 - Play important role cell differentiation
 - Determine of polarity of the cell
 - Microtubule control orientation of cellulose microfibril in plant cell

6.2. MICROFILAMENT:

- Visible E.M (ELECTRON MICROSCOPE) only.
- Discovered by Palevitz & co-worker 1974
- Found almost all eukaryotic cell
- Microfilament composed globular protein & Myosin
- Microfilament also called Actin filament
- Microfilament made up of Tubulin subunits.
- Microfilament- helical Polymer of dumbbell shaped subunits called Tubulin.

Functions of Microfilament

- Provide Cytoskeleton to support fluid matrix
- Intra cellular movement of particles organelles
- Help in Stream in movement of Cytoplasm.
- It help Locomotion-Microfilament gliding, Amoeboid motion
- Help movement of amoeba leukocyte & macrophage.
- Responsible for change in cell shape.
- MF responsible for Muscle contraction.
- MF cause movement of Ville during absorption
- MF responsible for movement of cell membrane Endocytosis & Exocytosis.
- Undulating movements by Microfilament (fibroblast to move)
- Helps formation of mitotic spindle fiber.

MICROTUBULES	MICROFILAMENT
• Hollow cylinder	• Solid rod
• About 200-270 Å thick	• 50-60Å thick
• 13 longitudinal	• Not composed protein filament
• Formed Protein tubulin	• Formed Protein actin & Myosin
• Submit dimmers bound GTP & GDP	• Submit are monomer bound ATP & ADP are contractive
• No contractive	• Are contractive
• No role Cytoplasmic Streaming Endocytosis	• Play a role Cytoplasmic Streaming endocytosis
• Exocytosis	• Exocytosis

- Intermediate Filaments (IF), Intermediate filament first described in 1968
- Supportive elements in Cytoplasm of Eukaryotic cell
- Absent-mammalian RBC and Prokaryotic cell
- Larger than Microfilaments (10nm thick), Made up non motile structure, Protein-Keratin-desmin-Vimentin
- It is formed -eight protofilaments.
- Keratin (IF) occur in epidermal cell help formation structure corner of dead cells

CELL & MOLECULAR GENETICS

- To not fibrils of Desmosomes made up bundle of Keratin
- Desmin - Muscle cell functional Unit capable of shortening
- Vimentin – interconnected nuclear membrane with cell membrane - Support Nucleus & maintain position of cells.
- Neurofibrils of nerve cell of bundle of co-polymerization of 3 distinct protein (NF_i, NF_m, NF_h)
- Glial cells have it composed of glial fibrillary acid protein (GFAP).

6.3. CENTRIOLE:

- Support fluid cytosol & maintain shape of the cell.
- Stabilize epithelia by bindings with Desmosomes.
- Form major structure protein of skin & hair, Muscle cell
- All components into functional unit of nerve it Provide strength to the axons.
- Keep Nuclear & other organelles in place.
- Cardiac muscle cells inter connected by Desmosomes help of function of heart.
- Centrioles (9+0) arrangements –Mitochondria -microtubule organism centre-or cell centre Van Benden 1887.
- centriole only seen both light and Electron microscope
- Location-all animal cell-motile plant cell Zoospores of algae-sperm cells of ferns-motile Algae
- A pair of centrioles called Diplosome located near the Nucleus
- Other organelles are called centrosome, kinoplasm, cytoplasm
- Centriole +centrospheres together called centrosome
- plant cells of bone marrow of a microtubule triplet-9+0 pattern centriole
- No membrane around the centrioles (membrane less cell organelles)
- 3microtubules of triplet often called tubules-each sub tubule is composed 13 parallel protofilament-Each Protofilament made of α & β tubulin dimer
- Peri centriolar satellites (or) pericentriolar matrix assemble of new centriole in G₂ phase of cell cycle.
- Centriole composed protein tubulin & lipids
- Centriole rich in ATPase enzyme

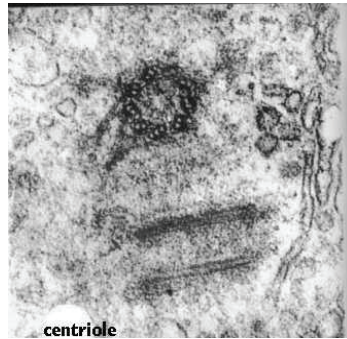


Figure 15: MICROSCOPIC VIEW OF CENTRIOLE

FUNCTION OF CENTRIOLES:

- Help organize spindle fiber & astral rays during cell division.
- Higher plant lack centriole and still form spindle fibers.
- Provide basal bodies of cilia & flagella
- Distal centrioles of sperm give rise to axial filament of tail.
- Basal Bodies (9+0 arrangement)
- Basal bodies also called basal granules.
- It occurs in Eukaryotic cell
- Basal bodies bear cilia & flagella
- Lie close to Plasma membrane
- Basal bodies otherwise called Kinetosome or blepharoplast.
- Basal bodies are same size chemical composition of centriole
- Basal bodies consist of nine microtubule triplets of tubulin (9+0 arrangement)
- Many animals and human- basal body of fertilized sperm flagellum enters egg-becomes centriole
- Give rise to cilia & flagella
- They act MT OC (microtubule -Organism centers)
- It helps anchor and movement of cilia & flagella
- Convert into centriole during cell division like Cilia & Flagella.
- Eukaryotic cells cilia & flagella fine Vibrative cytoplasmic processes
- Cilia & Flagella (9+0) arises from basal bodies
- Cilia found in Protozoan's class ciliate

CELL & MOLECULAR GENETICS

- Flame cells of flatworms epithelial lines of Respiratory tract
- Part of tubules in kidney eg: Renal tubules
- it present in Choanoflagellate of sponge, Astrodermal cells of coelenterate, Unicellular algae.
- Nematodes and arthropods lack cilia.
- Respiratory organ protect cilia without entry of dust particle into the lungs
- Tobacco smoke damage ciliated epithelium- allow dust to the burns-harm to lungs cause disease.

6.4. CILIA AND FLAGELLA



Figure 16: BACTERIA WITH FLAGELLA

- Flagella present in Protozoa called as Flagellate e.g. Choanocytes of sponges.
- Present in Gastro dermal cells of coelenterates ,Spermatozoa of most animals ,Lower plants ,Unicellular algae ,Zoospores of algae
- Centriole may form neat and right angle to pre existing one (Centriole cannot divide into two)
- Cilia and Flagella enclosed unit membrane of plasma membrane.
- Cilia covered membrane of Plasma membrane

- Cilia long-2-10 μ m
- Flagella long 150 μ m
- Cilia & Flagella axoneme composed 11 Microtubules (9+2 arrangement)

Chemical composition of cilia and Flagella:

- Microtubules composed globular unit of the protein tubulin.
- Arms of a microtubules contain ADP-ATPase enzyme-ATP – it help cilia Flagella movement- stroke & recovery stroke
- Interdoublet links are a Protein nexin are present.
- Cilia movement metachronous (or) Synchronous rhythms(cilia beat simultaneously)
- Ciliarybeat -power or effective stroke & recovery stroke (minimum resistance)
- Flagella beat independently Undulatory manner.
- Flagella two types –tinsel type Flagellum (stiff hair)-Himmer filament whiplasts flagellum.
Eg: Spermatozoa moves by Flagellar action.
- Microtubules-contractile protein tubulin-Ca²⁺, M+2, TP and calmodulin required assemble microtubules-Absent slime moulds amoeba prokaryotes-increase not during cell division.
- Microfilaments---Help in muscle contraction, solid rod, globular protein,(Actin), Filamentous protein (thick)(myosin), made up of monomer subunits, cyclosis or streaming movements, cleavage, Exo & endocytosis.
- Intermediate filaments----- made up of 8 Protofilaments, larger than microfilaments, present all eukaryotic cell except RBC,made up of keratin desmin, vimentin etc,maintain shape of the cell, transmission of contractile force.(micro circulatory system- microtubules circulatory of cell --- ER)
- Centriole inside microtubules are cart wheel structure.
- Centriole is (9+0) pattern made up of a microtubule triplets ring.
- Centriole is surrounding dense spheres called crown/massule/ pericentriolar satellite

CELL & MOLECULAR GENETICS

- Centriole – help high amount of ATPase enzyme production.
 - Centriole ----formation spindle fiber astral rays, give rise to cilia flagella, distal centriole of sperm give rise axial filaments.
 - Centriole called Basal bodies, Basal granules, Kinetosome, Blepharoplasts
 - It forms human sperm flagella. Help Enter the egg -Turn into centriole.
 - Basal bodies----plate like band present between basal body & shaft of cilia of flagella, C-sub fibre disappear additionally centre singlet fibre appear
 - Cilia and flagella are semi autonomous bodies
 - Cilia and flagella arises from basal body
 - Cilia and flagella are (9+2 arrangements)
 - Bacterial flagellum – single fibril composed protein flagellin. help
- Function of cilia and flagella
- Cilia and flagella act as organelles of locomotion in spermatozoa, many protozoan larvae certain animals' adults' animals.
 - Help captures food in many protozoan and some animals.
 - Cilia and flagella create water current (aquatic animals such as sponge, mussels tunicates, lancelets) ----- --Cilia and flagella water current bring food & O₂, remove waste matter & CO₂, remove gametes from the body.
 - Flagella circulate food in gastro vascular cavity of coelenterates.
 - Cilia of respiratory tract protect infection.
 - Urinary & genital tracts drive out urine and gametes.
 - Cilia thought to secrete sticky materials in ciliates.
 - Cilia probably sensory organelle and help in locomotion.
 - Cilia help capturing food, gases, remove waste.

6.5. PEROXISOME:

- Membrane bound micro bodies consist of crystalline core with dense matrix
- Contain enzyme peroxide biosynthesis
- Peroxisome – produce peroxide ----destroying and kill the bacteria
- Substrate(uric acid , amino acid) ----- H_2O_2 (Harmful to the cell) ----- $\text{H}_2\text{O} + \text{O}_2$

FUNCTION OF PEROXISOME:

- Very important role metabolism of sugar, amino acids, fats.
- Detoxify the alcohol in the liver by transferring H^+ to O_2 .

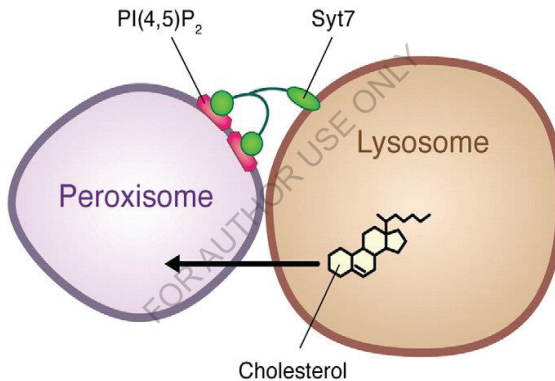


FIGURE 17: PEROXISOME WITH LYSOSOME

6.6. RIBOSOME:

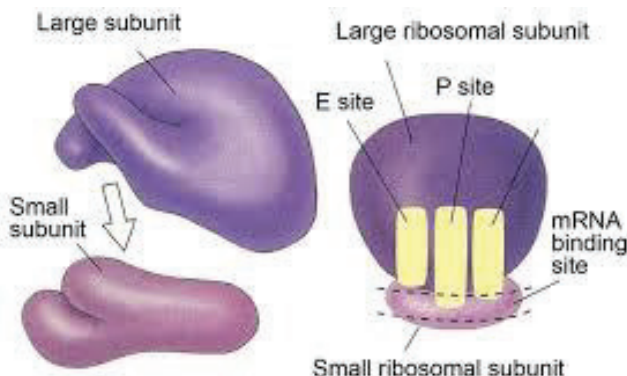


FIGURE 18: STRUCTURE OF RIBOSOME

- 70s ribosome, 50s (Mw 1800000), 30s (MW1000000).
- 80s ribosome-----60s (Mw3000000), 40s (MW1500000)
- Ribosome minute organelle seen only Electron Microscope.
- Discovered by Palade 1955.
- Ribosome otherwise called palade granules.
- Ribosome most abundant cytoplasmic organelles.
- Ribosome are protein & RNA (ribonucleic protein)
- 80% cell RNA are Ribosomal RNA (or) r RNA
- Ribosome non membranous cell organelle
- Prokaryotes ribosome found freely float in the cytoplasm
- Eukaryotes ribosome found cytoplasm lying free and attached with ER and Nuclear envelop-hence called cytoplasmic ribosome.
- Ribosome found inside cell organelles -----mitochondria (Mitochondria) and plastids of eukaryotes (chloroplast ribosome)—commonly called organelles ribosome's.
- Ribosome attached with ER help of ribophorin.
- Ribophorin substance present only in RER. Mass of ribosome expressed Svedberg unit Swedish scientist, t. Svedberg (1×10^{-13} cm |s|dyne|g. (Greater S Value = greater sedimentation).

CELL & MOLECULAR GENETICS

- 70s ribosome (20-30nm in diameter) – prokaryotic cell, mitochondria, plastids of eukaryotes.
- 80s ribosome – occur in cytoplasm, free or attached with RER Nuclear envelop, size 25-35nm in diameter.
- Ribosome larger subunit (dome shapes) Smaller subunit (ovoid shaped)
- Eukaryotes ribosome 80s 60s (larger) 40s (smaller)
- 5.85s 45 different protein 18S rRNA 33 ribosome protein 28S, 5S
- Prokaryotic 70s Ribosome (50s, 30s)
- Eukaryotic ribosome RNA (18S, 5.85S and 28S) Synthesized from Nucleolus
- Ribosome help protein Synthesized Cytoplasm.
- Nucleolus are called ribosome factory.
- Prokaryotes ribosome RNA & protein Synthesized Cytoplasm and assemble in Cytoplasm.
- ER bound Ribosome (RER) Large Subunit binds to membrane of ER Smaller subunit bound to large unit.
- Ribosome contain both 70s & 80s sub unit.
- Ribosome has 30 AA & 35 RNA nucleotide.
- Help growing accommodate molecule Polypeptide chain.
- Ribosome are Protein factories or work benches of proteins
- Mg^{++} ions needed bindings of two subunit of Ribosome
- Mg^{+2} form ionic bond with Phosphate group of rRNA of two subunit
- During Protein synthesis Smaller Unit of ribosome attach to mRNA- from Polyribosome or Polysome.
- Enzyme peptidyl transferase-responsible for formation of peptide chain
- Now (RNA responsible for peptide formation) proved by experiment.
- Ribosome + ER ---- synthesis protein ---- transported through ER
- Microsome differ from Ribosome.
- Ribosome abundant in Pancreatic cell, Endocrine gland, Lymphocyte, Yeast cell, Meristematic and Cancer cell
- Lower of ribosome-less active cell and starved cell.

Function of Ribosome

- Sites of protein synthesis
- Site for the attachment mRNA & tRNA
- Tunnel of larger subunit protect polypeptide till attain secondary structure.

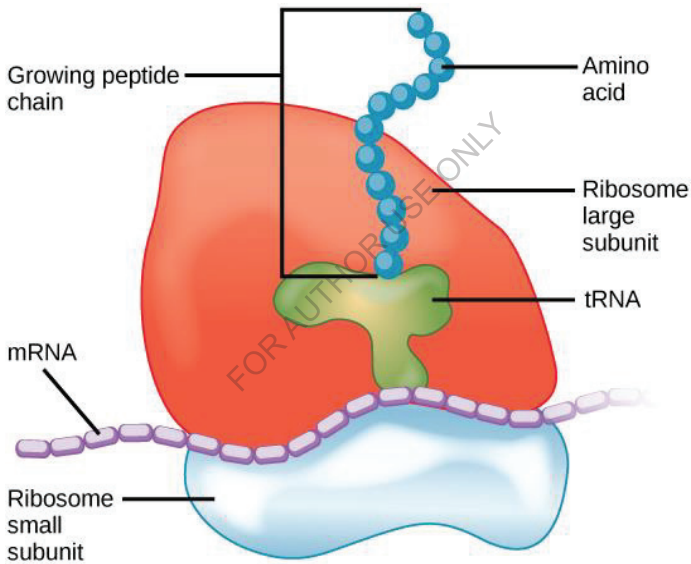


FIGURE 19: FUNCTION OF RIBOSOME

Chapter 7

Structure of DNA and RNA

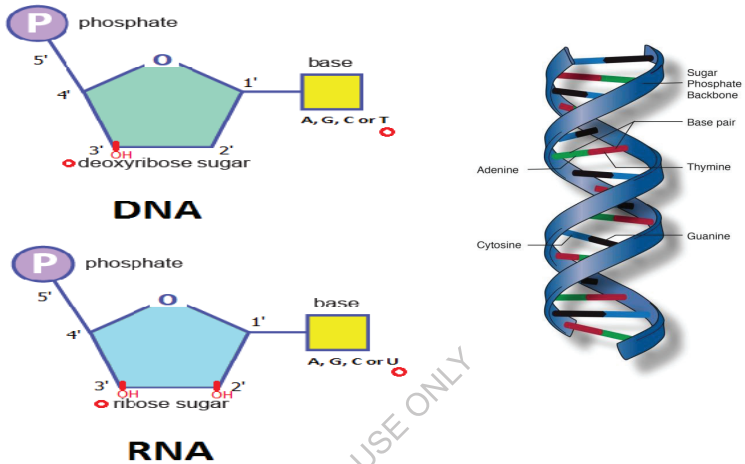


Figure 20 : DNA and RNA DOUBLE HELIX MODEL

STRUCTURE OF DNA:

- DNA basis unit Deoxyribonucleotide
- Deoxyribonucleotide
- Nucleotide –Nitrogen base weakly basis- Hydrophobic compound
- Types of Nitrogen bases- -Purine and Pyrimidine Nucleotides
- Purine – Adenine(A)Guanine(G), Urasil(U)
- Pyrimidine- Cytosine(C) and Thiamine(T)
- Deoxyrobose Sugar in DNA
- C5 H10 O4-pentose ring-102-4 Carbon atoms
- H3 po4 of DNA(Phosphoric acid) Acidity to Nucleic acid
- Pentose Sugar & Purine and Pyrimidine combine with β -N-Glycosidic linkge .
- C-N-C Nucleotide
- Nucleoside + Phosphoric acid-Nucleotide

CELL & MOLECULAR GENETICS

- Phosphate attached with pentose sugar (5th or 3rd) position by Phosphodiester bond.
- Example of higher nucleotide in free state ATA, TTP, GTP, ADP, GTP, GDP, CDP, TDP
- Backbone of DNA molecules sugar and Phosphate
- DNA helix one stand 5'-3' another stand 3'-5'.
- DNA stand antiparalleled direction connected with Hydrogen bond. Hydrogen bond between Nitrogen bases A=T, G=C
- Watson & Crick model of DNA
- DNA double helix 20 Å
- 1 turn 34 Å (3.4 nm)
- 1 turn contain 10 Base pair
- Major Groove 22 Å (2.2 nm)
- Minor Groove 12 Å (1.2 nm)
- Both Chain anti parallel in direction.

DIFFERENCE BETWEEN PURINE AND PYRIMIDINE NUCLEOTIDES:

PURINES (A,G,U) (Nucleotide)	PYRIMIDINES (C,T) (Nucleotide)
• Large sized	• Small sized
• Double ring	• Single ring
• Pyrimidin ring	• Imidazole ring
• 9 Membered ring	• 6 Membered ring
• Nitrogen is at position 1,3,7 & 9	• Nitrogen is at position 1 & 3
• 9th position, Nitrogen combine with pentose sugar	• 1st position Nitrogen combine with Pentose sugar
• Adenine A, guanine G	• Thymine (T), Cytosine (C), Uracil (only RNA).

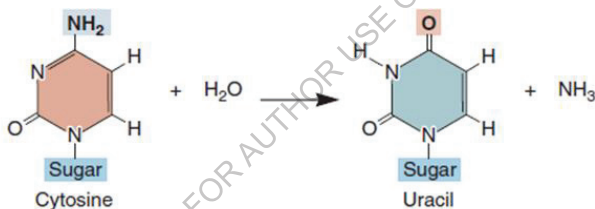
Different forms of DNA:

	A-Form	B-Form	C-Form	Z-Form
Abbreviation	A-DNA	B-DNA	C-DNA	Z-DNA
Structure proposed by	-	Watson & crick	-	Rich et. al. 1979
Base pair/turn	11 BP	10 BP	28/3 BP 9.33	12 BP
Axial rise of each base pair	2.5A ⁰	3.37 A ⁰	3.32 A ⁰	3.32 A ⁰
Tilt of base pair	20-2 (Tilted)	6.3perpendicular	-7.80	+60
Pitch of helix	28.15 A ⁰	34 A ⁰ /33.37	31A ⁰	
Stability	Metastable A-D	Metabollically stable	-	-
RNA & DNA hybrid	possible	-	-	-

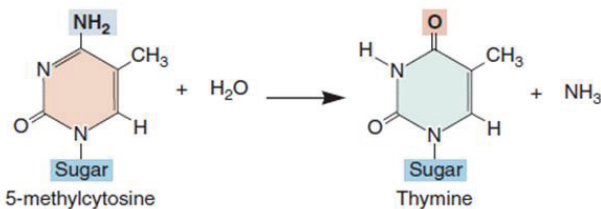
Forms	Sugar Pucterin Carbon	No. of. Bp Per Turn	Axial Rise	Length	Base Pair Tilt	Diameter of Molecule
A-DNA	31- endo	11	2.56A ⁰	28.16 A ⁰	+20.20	23A ⁰
B-DNA	31-exo	10		3.37 A ⁰		20 A ⁰
C-DNA	31- endo	9.33		3.37 A ⁰		19 A ⁰
D-DNA	31-exo	8		24.24 A ⁰	-16.70	-

- DNA right handed helix - B-DNA
- DNA left handed form in 1979 Rick et al.- Z- DNA
- Palindrome DNA same sequence of Nucleotide in opposite direction.

- ACGT } it form Cruciform loops
- TGCA }
- Nucleotide repeated many time is called repetitive DNA
- Single stranded DNA e.g.: Bacteriophage Virus 174
- During replication single stranded
- DNA become double Stranded
- Bacteriophage & animal Virus contain Single rounded DNA
- Circular DNA -double helix DNA
- Most phase DNA form of circle
- Example circular DNA-Qx174 Phases-Polyoma viruses-mitochondria DNA.
- RNA-polynucleotide but Pentose ribose tree
- Double stranded RNA also present in echoviruses & wound tumors viruses.
- Single Stranded RNA - highly folded help in stability of the molecules



(a) Deamination of cytosine



(b) Deamination of 5-methylcytosine

Figure 26: MOLECULAR STRUCTURE OF DNA NUCLEOTIDES

7.1 Types of RNA

TYPES	rRNA	mRNA	t RNA
Percentage of Total RNA of The Cell	80%	3-5%	10-20%
Sedimentation Co Efficient	28s, 18s, 5.8s, 5s, 23s, 16s, 5s	8s	3.8s
No. of Nucleotide	5s RNA-120 Nucleotide-16s-18s RNA(1600-25000)Nucleotide 23-28s RNA (3200-5500)	E.Coli 900-15000 Nucleotide	73-93
Molecular Weight	23s(11x106), 30s(0.55x106)	50000	2500-30000
Unusual Bases	Small Amount of Methylated Bases	Small Amount of Unusual Bases	High Content of Unusual Bases
Site of Synthesis	Derived From (Nuclear DNA)	Synthesized Nuclear DNA	Synthesized Nuclear DNA
Beginning of Synthesis	Synthesis Begin At Gastrulation	Synthesis Early Cleavage Stage(mRNA Found in Ovum)	Synthesis End of Cleavage DNA
Base of Relationship DNA	rRNA Formed From Small Section Of DNA	Formed From All Section Of DNA	tRNA Formed From Section of DNA
Function	mRNA And tRNA Bind With Ribosome	Convey Genetic Information DNA- Ribosome	Attached AA To mRNA Template

- Messenger RNA(mRNA) (or) template RNA

CELL & MOLECULAR GENETICS

- Ribosome RNA (OR) r RNA
- Soluble RNA (OR) s RNA or transfer RNA or tRNA
tRNA.
- Most Of The DNA Is Found In Chromosome.
- Some DNA Found In The Cytoplasm, Mitochondria And Chloroplast.
- mRNA Found Nucleolus And Cytoplasm, rRNA And tRNA Formed In Cytoplasm.
- DNA-----Replication -----DNA Formed
- DNA-----Transcription -----RNA Formed
- RNA Does Not Replicate And Transcription.
- DNA---- 4.3 Million Nucleotide.
- RNA ----12000 Nucleotide Only

• Properties	• A 11bp	• B 10bp	• Z 12 Bp
• Helix Sense	• Right	• Right	• Left Handed
• Diameter	• 26Å ⁰	• 20Å ⁰	• 18Å ⁰

- Right handed for B-DNA-----Positive
- Left handed for B- DNA----Negative
- Polidrome sequence DNA ----Hairpin (OR) crucifor DNA
- Chromosome first discovered by 1842 swiss botamist Karl Wilhelm von Nageli
- Chromosome ---- coloured bodies ----van beneden 1888
- Thread like chromosome E. Straburger (1975).

Chapter 8

DNA Replication

- DNA replication occurs S-phase of cell cycle.
- Replication takes place 5', -----3', end.
- Replication starts at particular point called origin of replication /origin
- DNA unpairing rich in A=T base pair
- Unidirectional replication
- (viruses, mitochondria, plastids, bacteria)
- Bidirectional replication (Eukaryotic replication)
- Single replication unit is called replication
- Neuron theory –Waldeyer
- Uncoiling of DNA-----helps the enzyme (helicase (or) unwinding)
- Replication fork reduced by the enzyme in Eukaryotes ---topoisomerase
- Replication fork reduced by the enzyme in Prokaryotes--- gyrase
- Activation of nucleotides called Deoxyribonucleotide found in Nucleoplasm----- phosphorylated by phosphorylase enzyme.
- Example of phosphorylated nucleotide-- dATP,dGTP,dTTP, dCTP.
- DNA replication initiated by RNA Primer
- RNA Primer made up of minimum 5 nucleotide
- RNA synthesized at 5' end presence of primase.
- Editing of DNA strand by DNA polymerase I bounded by Ligase
- Extra phosphate group separated from nucleotide by enzyme
- Formation replication DNA strand by DNA polymerase III (Zipper duplication).
- DNA strand having 5' end to form leading strand
- Newly formed small segments on the DNA strand is called Okazaki fragments (30000-40000 Nucleotides)-----joined to form new strand is called lagging strand
- Meselson & Stahl experiments 1958, Meselson & Stahl experimentally proved semi conservative replication of DNA.
- Labelled DNA with help of N14 or N15 (15 NH₄CL)
- One strand conserved while second newly formed --- semi conservative replication of DNA.
- Replication takes place during inter phase between two mitotic cycles .

CELL & MOLECULAR GENETICS

- DNA Replication requires -----RNA Polymerase, DNA Template, a primer, Deoxyribonucleotide triphosphates (d ATP, d GTP, d TTP, d CTP), Polynucleotide ligase Mg^{2+} , DNA unwinding protein, super helix, relaxing protein.
- Replication starts with incision made by incision enzyme (Endonuclease).
- Unwinding of strands – super helix relaxing protein.
- Initiation of DNA synthesis requires RNA Primer.

RNA Primer synthesized by the DNA template with help of RNA polymerase

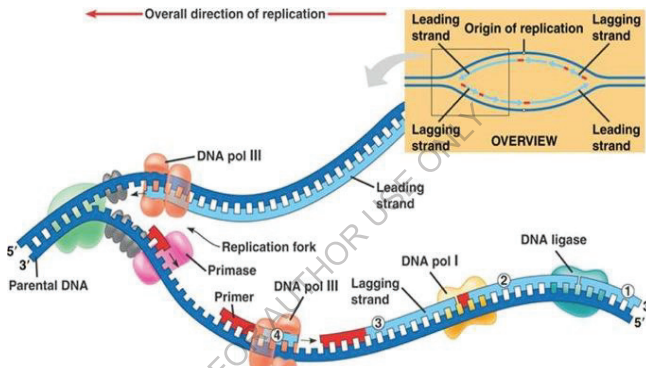


Figure 27 : GRAPHICAL IMAGE OF DNA REPLICATION

- DNA New strand from direction of 5', -----3',
- Watson and crick explained DNA replication.
- Watson and crick model of DNA (B DNA) -----conservation, semi conservation, dispersive replication.
- Semi conservative replication of DNA proposed by Watson and Crick (1953).
- Semi conservative replication experimentally proved by Measelson and Stahl.
- Semi conservative replication demonstrated by autoradiography Taylor et.al (1957).
- Radioactive N14-light + N15- heavy

- DNA photography taken basis of X-ray diffraction studies by Wilkins and Franklin.
- Erwin Chargaff rule 1949
 - a. A-T
 - b. G-C
 - c. $[A]+[G] \text{ Purine}=[T]+[C] \text{ Pyrimidine}$
 - d. $[A]=[T] \neq [G]=[C]$
- Bacteria, viruses contain single stranded DNA.
- Bacterial chromosome contain double stranded DNA ring.
- Replication of DNA during mating of bacteria special DNA synthesis with help of replication during oogenesis e.g. *Xenopus* larva.
- In vitro synthesize biologically active DNA made by Arthur Kornberg.
- DNA polymerase enzyme discovered by Kornberg 1957.
- DNA polymerase responsible for polymerization of Deoxyribonucleotide tri phosphate on DNA template.
- DNA polymerase enzyme catalyzed by four Ribonucleoside tri phosphates---d ATP, d TTP, d CTP.
- Prokaryotic replication---DNA Polymerase I, DNA Polymerase II, DNA Polymerase III.
- Eukaryotic replication--- Polymerase α , Polymerase β , Polymerase γ .
- Editing enzyme: Polymerase I.
- Polymerase I repair the DNA
- DNA polymerase I with Zn so called metallo enzyme.
- About 400 molecular of polymerase I present in *E.coli*
- DNA template polymerase I attached single stranded region of DNA.
- DNA template near double stranded region polymerase I & II.
- Removable of nucleotide (hydrolyzing DNA) takes place 3',----- 5',
- Hydrolyse occur in 5', -----3', direction.

8.1 DNA REPAIR:

- Genes are stable molecule it may change by a process of mutation.
- Ionizing radiation ----- X-rays, gamma rays, cosmic rays.
- Non ionizing radiation – UV light
- UV rays potent mutagen for unicellular organisms.
- Max absorption of UV by DNA at wavelength of 254nm.
- Mutagenic effect of UV rays----formation of thymine dimers-----
Thymine dimers attached with Adenine
- Dimer repair by---photo reactivation, Excision repair, Recombination repair.
- Enzymatic cleavage of thymidine dimer by visible light e.g. PR enzyme (or) photolyase
- Excision repair occur efficiently in dark & presence of blue light.
- Endonuclease makes cut in the backbone of DNA.
- Polymerase I act as a primer.
- Synthesis of new strand with help of DNA ligase enzyme.
- After DNA duplication it is called Post replication repair.

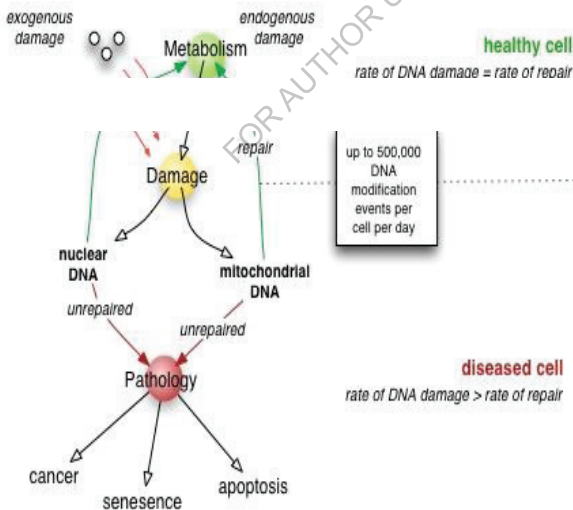


FIGURE28: DNA REPAIR

Chapter 9

Amino Acid

- Aminoacid – Asparagine- glutamine
- Secondary AA-Proline-Hydroxyproline

9.1 DERIVED AMINO ACID:

1. B-Alanine
2. Phosphoserine & Phosphothreonine(Enzyme)
3. Ergothionine(blood)
4. Hydroxylysine(collage)
5. Ornithine(uses biosynthesis)
6. Citrulline(uses Biosynthesis)
7. Homocysteine(methionine)
8. Cystathionine(methionine met)
9. V- aminobutyric acid
10. β - Aminobutyric acid (thymine metabolism)
11. Dihydroxy Phenyl
12. Cysteine
13. Taurine
14. Hydroxy Proline
15. Thyroxine & Thyroidosis.

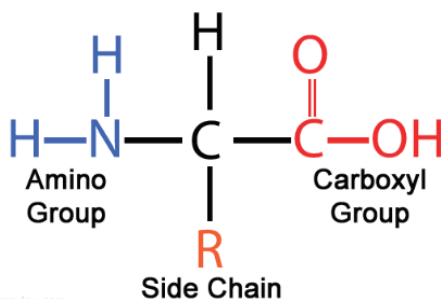
Amino Acid Structure

Figure 29: STRUCTURE OF AMINO ACID

9.2 Some animal's protein and their Molecular Weight

- Insulin-12,000 in Daltons
- Cytochrome(Horse Heart) 12100
- Trypsin-23800
- Pepsin-35500
- Oval albumin 44000
- Serum albumin (Human) 65000
- Haemoglobin(Human)67000
- Y- Globulin(Human)
- Catalase 250000
- Collase 345000
- Tryolobin(pi) 650000
- Simple protein – albumin-globuline-histones-Prolamine- Protamine
- Conjugated Protein- Glycoprotein & mucoprotein ex: Musin of saliuva
- Lipoprotein ex: Lipo Vitelhine egg yolk serum
- Protein of brain nerve tissue
- Nucleoproteins-Viruses & NA protein
- Chromo proteins
- Hemoglobin, Haemocyamine- Havoprotein, Cytotochorame
- Phosphoprotein casein in milk
- Metalloprotein
- Carbonic an hydrase enzyme (zinc)
- Collagnes-Fibrous Protein of Corrective tissue
- Globular Protein –enzyme – Hormone-immunoloical Protein
- Collagens boiled Elastin.
- Quart man protein structure called highest level of protein Structure.

9.3. TYPES OF ENZYMES:

1. Oxidoreductase- Oxidation catalysis & reduction: Ex. Oxidases, hydronase, Oxygenase, Reoxidase.
2. Transferase-Transfer of any group from one Molecule to other. Ex: Transferase
3. Hydrolases-Complex molecule into two compounds: Enzyme act on ester-P-N bonds-C-C halide-C-N bonds acid-Peptide anhydride-Ether-glycosyl
Eg. Proteases, esterase, Phosphatase, Nucleases, Phospholipases, digestive enzyme.
4. Lysozymes-add or remove group from Chemical compound. Acts on C-C, C-O, C-N, C-S, C-halide bonds.
5. Isomerase- Isomerization or intra molecular rearrangement.
Ex: Intramolecular Oxidoreductase, Intra Molecular transferase, Intermolecular Lysozymes, isomerase, epimerases.
6. Ligases of Synthetases- Linkage of the molecule by split phosphate bond C-O, C-S, C-N, and C-C BOND.

Enzyme Specific action

PH

Temp

Affect enzyme action.

7. Zymogens- Catalytic activity (Inactive form)
8. Trypsinogen---Activated by enterokinase (Inactive form) (Intestine enzyme) Pancreatic cells.
9. Pepsinogen--- activated by Parietal cells(Chief cells of stomach)
Protein contain-organic – inorganic unit + Prosthetic group.
Ex: Cytochrome +Metalloporphyrine complex. Co enzyme- enzyme+ small molecules of other chemical-activates
10. Apo enzyme- inactive enzyme:Apo enzyme & co enzyme called as Holoenzyme E.g hydroxyenes+ co enzyme NAD NADP-Activated functions
11. Similar activity &almost similar molecular structure- Iso enzyme
100 (isoenzyme in the cell) Ex: Lactic dehydrogenase LDH.

Chapter 10

Genetics and Genetic Engineering

10.1. MUTATIONS AND GENETIC CODE:

Sudden changes of structure of Nucleic acid.

1. Point Mutation

- It involves a change in single base pair.
- An example of point mutation is a change of single base pair in the gene for beta globin chain that results in glutamine in place of valine.
- It causes a disease called sickle-cell Anemia.

2. Frame-Shift Mutation

- It occurs where addition/insertion or deletion of one or two bases changes the reading from the site of mutation, resulting in a protein with a different set of amino acids.
- Insertion or deletion of three or its multiples of bases do not alter the reading frame, but one/more amino acids are coded in the protein translated altered.

3. Silent Mutation

- It occurs, if a base change in a codon does not alter the amino acid coded.
- One Gene-One Enzyme Hypothesis
- 'One gene-one enzyme hypothesis' was proposed by Beadle and Tatum (1941), according to this, particular gene controls the synthesis of specific enzyme, Beadle and Tatum conducted experiment on pink bread mould (*Neurospora crassa*).
- *Neurospora* is an Ascomycete fungus, which is widely used in genetic research due to the following characters
 - It is easy to grow
 - It can be bred in large number.
 - It has a very short life cycle that is about 10 days.
 - It has haploid vegetative stage.

- Beadle and Tatum stated that each gene has the information to produce one enzyme.
- Now, it is known that one gene-one enzyme hypothesis is not completely valid because some regions of DNA produce tRNA or rRNA rather than mRNA (which produces a protein).
- Therefore, some genes do not code for protein, instead, they specify RNA components of the translational apparatus.
- Yanofsky et. Al (1965), proposed a modified hypothesis known as one gene-one polypeptide hypothesis because he observed that the enzyme tryptophan synthetase of E.coli consists of two separate polypeptides.
- Yanofsky, Brenner and his colleagues established colinearity between genes and polypeptide products.
- They concluded that the linear sequence of nucleotides in a gene determines the linear sequence of amino acids in a protein.

10.2 .VARIOUS STEPS OF PROTEIN SYNTHESIS:

- Amino acid activation and involvement of rRNA. tRNA reads the message coded on mRNA.
- The code in the form of three bases is called codon. tRNA are present in the cytoplasm.
- The two dimensional structure of the tRNA is clover leaf like, but the three dimensional form is L-shaped. tRNA is activated by amino acyl-AMP enzyme complex and specific amino acid is attached to 3' end of tRNA molecule.
- Activation of amino acid is catalysed by the amino acyl-tRNA synthetase enzyme.
- tRNA is specific for a particular amino acid. tRNA has one loop called anticodon loop.
- Anticodon has complementary base of the message on mRNA.
If AUG is the code of mRNA, the anticodon on tRNA will be UAC.

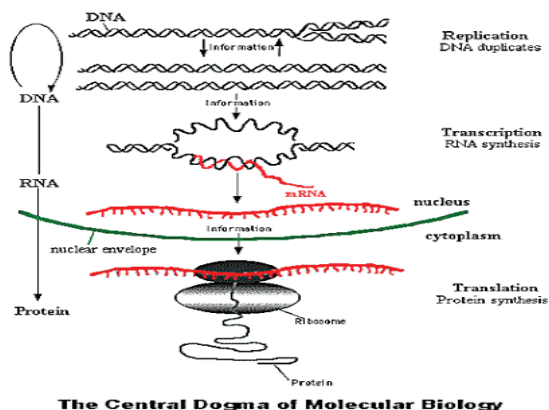


FIGURE 30: CENTRAL DOGMA OF MOLLECULAR BIOLOGY

10.3. INITIATION OF PEPTIDE BOND SEQUENCE

- mRNA binds to small unit of ribosome in the presence of initiation factor.
- Protein synthesis mostly starts with methionine (formylated methionine in case of prokaryotes).
- The tRNA first binds to P-site of ribosome. Then the large subunit of ribosome gets attached to the small subunit forming a complex.

10.4. tRNA THE ADAPTER MOLECULE

- tRNA is also called the transfer RNA. tRNA has an anticodon loop that has bases complementary to the code, and it also has an amino acid acceptor end to which it binds to amino acids, tRNAs are specific for each amino acid.
- For initiation, there is another specific tRNA that is referred to as initiator tRNA. There are no tRNAs for stop codons.

10.5. ELONGATION

- Second tRNA with amino forms hydrogen bonds with the second codon on mRNA.
- This tRNA comes on site 'A' of ribosome.

- Peptide bond is formed between the first amino acid and second amino acid. The enzyme peptidyl transferase catalyses the reaction.
- Now the second amino acid shifts to 'P' site.
- The process of shifting of amino acid from "A" site to 'P' site is called translocation.

10.6. TERMINATION

- When a code UAA, UAG or UGA comes, it is the indication of termination. UAA, UAG and UGA are also called as non-sense codons.
- The process of initiation, elongation and termination involve the mediation of several factors and GTP for energy.

10.7. TRANSLATION

- The decoding of the message of mRNA in the form of protein is purely unidirectional process called translation.
- It takes place on ribosomes.
- Different proteins do different things; structural proteins determine the organism's physical form, functional proteins determine how the cells, tissues, organs and indeed the whole organism survives in its environment and regulatory proteins act as switches, where appropriate to direct metabolic responses as needed.
- Ribosomes are made up of rRNA and protein. It consist of two parts large and small.
- mRNA moves through the groove between the large and small subunits of ribosomes.

10.8. STEPS OF TRANSLATION PROCESS:

- mRNA binds to the two subunits of a ribosome.
- Initiator codon on mRNA (AUG) binds to first
- Anticodon on the mRNA (UAC). This tRNA carries the amino acid methionine.
- The next codon on the mRNA determines the next tRNA anticodon, which binds with it and so, the next amino acid in the polypeptide.

CELL & MOLECULAR GENETICS

- A peptide bond forms between the amino acids.
- The tRNA detaches from the ribosome and returns to the
- Cytoplasm to pick up another amino acid.
- The process continues, with codons matched by
- Appropriate tRNA anticodons and peptide bonds forming between the amino acids.
- Eventually a stop codon is reached on the mRNA.
- The ribosome releases the mRNA and the newly formed
- Polypeptide (further processing of the polypeptide may Occur to make it functional).

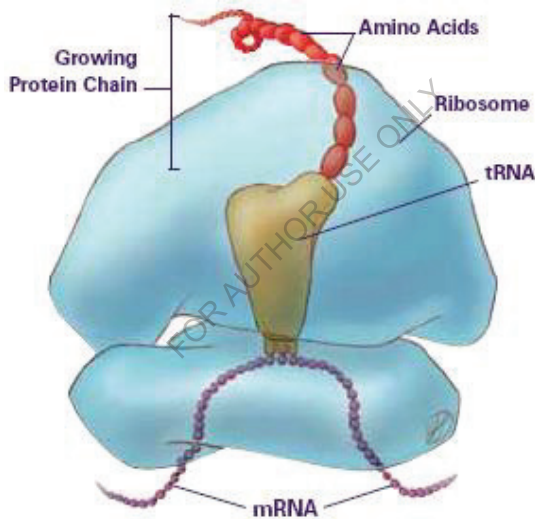


FIGURE 31. GRAPHICAL IMAGE OF PROTEIN SYNTHESIS

Chapter 11

Gene Expression**11.1. GENE EXPRESSION:**

- Gene expression is the process by which information contained in genes is decoded to produce other molecules that determine the phenotypic traits of organisms.
- The process is initiated, when the information contained in the base sequence of DNA is copied into a molecule of RNA is used to determine the linear order of amino acids in a polypeptide chain.
- The detail process of gene expression including transcription and translation is described earlier.
- The processed RNA molecule is used to specify the order in which amino acids are joined together to form a polypeptide chain.
- In this manner, the amino acid sequence in a polypeptide is a direct consequence of the base sequence from an RNA base sequence is called translation and the protein made is called gene product.

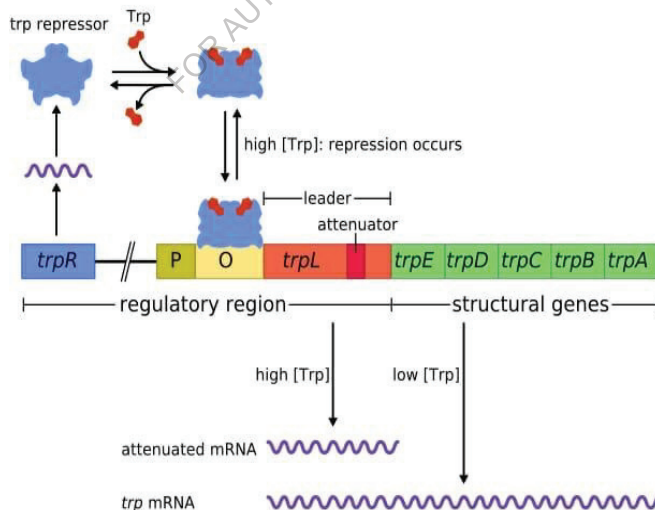


FIGURE32: GENE EXPRESTION

11.2. GENE REGULATION:

Gene regulation is the mechanism by which the expression of different genes controlled.

11.3. Gene Regulation or Regulation of Gene Expression Takes Place at the Following Steps.

1. Gene amplification, destruction or distribution
 2. Transcription
 3. Post transcription
 4. Translation
 5. Post translation
- In prokaryotes (e.g., bacteria), genes are organised into operon. Operon is a group of structural genes, whose transcription is regulated by a set of a regulator gene, a promoter and an operator, which are often located upstream or onto the left of the structural genes.
 - Two French scientists, Jacob and Monod (1961) proposed an operon model for regulation of gene activity in prokaryotes.
 - Operon is a part of genetic material (or DNA). Which acts as a single regulated unit having one or more structural genes, an operator gene, a promoter gene, a regulator gene, a repressor and an inducer or corepressor (from outside)
 - Repressor is a proteinaceous substance synthesised by the regulator gene, which blocks the operator gene, so that the structural genes are unable to form mRNA by transcription and protein (enzyme) formation is halted.
 - Inducer is a chemical substance (substrate, hormone or other metabolite), which binds to repressor and changes the repressor into non-DNA bind state so as to free the operator gene.
 - Lactose acts as inducer in *Iac* operon of *E.coli*.
 - Regulator gene controls the activity of operator gene by producing repressor protein, which gets attached to operator gene and checking mRNA synthesis (transcription is stopped).
 - Promotor is that sequence of an operon, where RNA polymerase binds and initiates transcription.

- Operator gene controls the activity of structural genes taking part in the synthesis of proteins.
- Operator gene is situated between the promoter gene and first structural gene, where it serves as binding site for repressor protein.

The control of protein synthesis by regulator proteins may be inducible or repressible.

11.4. LAC OPERON

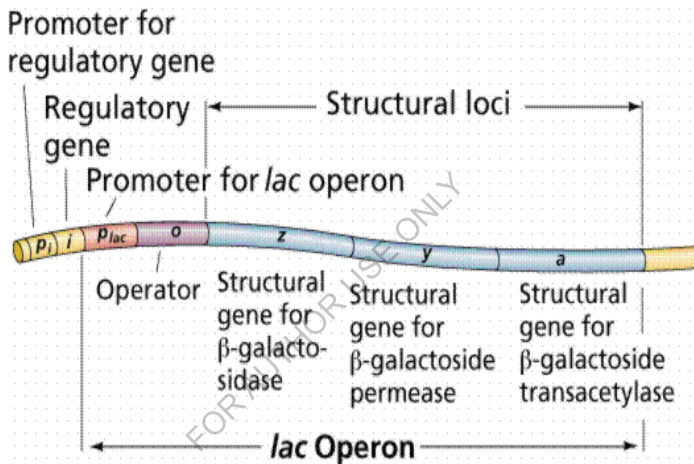


FIGURE:33 LAC OPERON

- Lac operon is an inducible operon, in which presence of key metabolic substance, i.e., lactose induces transcription of structural genes (or active repressor + inducer is equal to inactive repressor).
- The lac operon of E.coli.
 - a. Arrangement of genes in lac operon.
 - b. The repressor produced by regulator gene is in an active form and binds to the operator DNA preventing transcription initiation.
- But in the presence of inducer, the repressor molecules are inactivated by the inducer molecules so, that they can no longer bind

to the operator DNA. As a result, the structural genes are transcribed.

- The system of regulation in lac operon is a negative control as interaction of DNA with repressor protein inhibits gene expression or operon is on, but is kept off by regulator gene.
- However, lac operon is also under positive control through Catabolite Gene Activator (CGA protein) or Cyclic-AMP receptor Protein (CAP).
- CAP the structure found at the 5' end of many eukaryotic mRNAs. It consists of 7-methyl-guanosine-pppX, where X is the first nucleotide encoded in the DNA.
- It is not present in prokaryotic mRNAs; it is added post transcriptionally near the TATA (Hogness) box.
- CAP not to be confused with cap. CAP is catabolite gene activator protein (sometimes CRP or CGA); it participated in the initiation of transcription in prokaryotes.
- In positive control, transcription of the concerned operon does not take place unless the regular protein is bound to the operator; such regulator proteins are called activators.
- Negative regulation is widespread in prokaryotes (e.g. lac operon, trp operon) and lower eukaryotes, while positive regulation predominates in eukaryotes.

11.5. Trp Operon

- The tryptophan operon (trp operon) in bacteria is a repressible operon, where repressor is inactive and it becomes active as DNA binding protein complexed with corepressor (i.e., tryptophan).
- In absence of tryptophan, the operator site is open to binding by RNA polymerase, which transcribes the structural genes of trp operon.

11.6. COMPARISON BETWEEN INDUCIBLE AND REPRESSIBLE OPERONS

FEATURE	INDUCIBLE	REPRESSIBLE OPERON
1. Regulator protein (negative state)	Active (binds to operator)	Inactive (does not bind to operator)
2. Regulator protein after association with effector	Becomes inactive	Becomes active
3. Transcription of the operon in the presence of effector	Occur	Does not occur
4. Effector termed as	Inducer	Co repressor
5. Active operon is termed as	Induced	De repressed
6. An operon that cannot become active is called	Uninducible	Super-repressed

11.7. COMPARISON BETWEEN NEGATIVE AND POSITIVE CONTROL OF TRANSCRIPTION

FEATURE	NEGATIVE CONTROL	POSITIVE CONTROL
Default state of operon	Transcribed	Not transcribed
Interaction or regulator protein with operator / promoter	Prevents transcription	Initiates transcription
Inactive regulator protein produces	Recessive constitutivity	Recessive Uninducibility or super-repression
Defective operator/DNA binding site produces	Dominant constitutivity	Dominant super-repression/ Uninducibility

11.8. HUMAN GENOME PROJECT (HGP)

- HGP was a 13 year project that was launched in the year 1990 and completed in 2003.
- The project was co-ordinated by US, Department of Energy and the National Institute of Health.

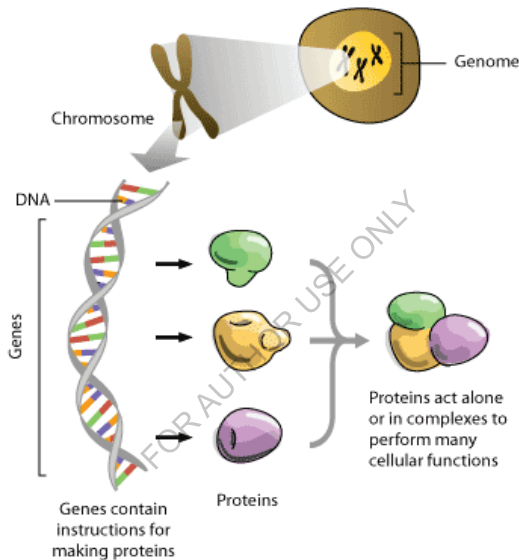


FIGURE 34: HUMAN GENOME PROJECT

GOALS OF HGP

- Identify all the genes in human DNA.
- Determine the sequence of the three billion base pairs present in human DNA.
- Store the information in databases.
- Improve the tools for data analysis.
- Transfer the technologies to other sectors.
- Address the ethical, legal and social issues that may arise from this project.

Methodologies of HGP

- Expressed Sequence Tags (ESTs) approach focuses on identifying all the genes that expressed as RNA.
- Sequence Annotation approach focuses to simply sequence the whole set of genome, that included all the coding and non-coding sequences.

Advantages of HGP

- Knowledge of the effects of variations of DNA among individuals can provide the ways to diagnose, treat and even prevent a number of diseases/disorders that affect human beings.
- It provides clues to the understanding of human biology.

11.9 .DNA FINGERPRINTING:

The DNA fingerprinting technique was developed to find out markers for inherited diseases. DNA fingerprinting involves.

- Isolation of DNA.
- Digestion of DNA by restriction endonucleases.
- Separation of DNA fragments by electrophoresis.
- Hybridisation using labeled VNTR probe and
- Detection of hybridized DNA fragments by autoradiography.



Chapter 12

Human Genome

The salient features considered in human genome are listed here.

12. HUMAN GENOME:

- The human genome contains 3164.7 million nucleotides (base pairs).
- The size of genes varies.
- A total number of genes is estimated at 30000 and 99.9% of the nucleotides are the same in all humans.
- Only less than of the genome codes for proteins.
- Repetitive sequences make up a large portion of the human genome.
- Scientists have identified about 1.4 million locations, where DNA differs in single base in human beings. These are called single Nucleotide Polymorphisms (SNPs).

12.1.SOME ORGANISMS AND THEIR CHROMOSOMES NUMBERS

Organism	Haploid Number Of Chromosome	Linkage Group
Drosophila	N=4	4
Garden pea (pisum sativum)	N=7	7
Maize (Zea mays)	N=10	10
Human (Homo sapiens)	N=23	23

HEREDITY:

- Heredity like begets like) is the transmission of characteristics from parent to off springs.
- Such transmissible characters are called hereditary characters.
- Hereditary traits or characters are specified by gene, which are integral parts of chromosomes.

VARIATIONS:

- These are differences in morphological, physiological, cytological and behavioural characters shown by the individuals of the same species, race and family.
- The main sources of variations are crossing over, chance combination of chromosomes during meiosis, fertilization and mutations.
- Mutation is the ultimate source of variations.

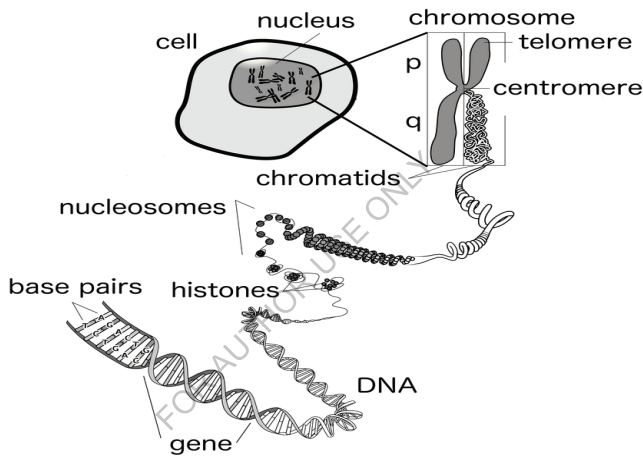


FIGURE 35: HUMAN GENOME TECHNOLOGY

12.2.DIFFERENT GENETIC TERMINOLOGY:

- Mendelian Factor Mendelian factor now known as gene (Johannsn; 1905) is the unit of inheritance, which determines biological character of an organism and passes from one generation to next through gamete.
- Chemically gene is a linear segment of DNA.
- Alleles or Allelomorphs Formerly, the term allele was used for the factors representing the two alternate terms of a trait., e.g., tallness and dwariness in case of height in pea (T and t) However, presently the term allele is used for any two forms of a gene present on the same locus in the two homologous chromosomes.

- **Gene locus** A particular region of part of chromosome representing a single gene is called gene locus.
- **Dominant Allele** Out of a pair of alleles, allele which can express itself phenotypically whether present in homozygous or heterozygous (hybrid) state is known as dominant allele.
- It is denoted by capital letter. E.g., R for red color.
- **Recessive Allele** Contrary to dominant allele, recessive allele cannot express itself phenotypically, in the presence of its contrasting factor heterozygous condition, e.g., 'r'

12.3. DIFFERENCE BETWEEN DOMINANT AND RECESSIVE GENES

Dominant Gene / Allele	Recessive Gene / Allele
1. Dominant gene is able to express itself even in the presence of its recessive allele.	1. Recessive gene or allele is unable to express its effect in the presence of dominant gene allele.
2. Dominant gene can form complete polypeptide or enzyme for expressing its effect. E.g., color of flower, yellow color of seed.	2. The recessive allele forms an incomplete defective polypeptide or enzyme.

12.4. DIFFERENCE BETWEEN DOMINANT AND RECESSIVE CHARACTERS

DOMINANT CHARACTER	RECESSIVE CHARACTER
1. The character, which is expressed in the hybrid is called dominant character	1. The character which remains suppressed in the hybrid is called a recessive character.
2. In Mendel's monohybrid cross between a tall pea plant and a dwarf pea plant, the hybrids were all tall (Tt) i.e., tallness (T) was dominant.	2. In the same experiment, the dwarf character was suppressed in F ₁ hybrids (Tt) hence, dwarfness (t) was recessive.
3. A dominant character is denoted by a dominant gene (T) a pair of dominant alleles (TT)	3. A recessive character is denoted by a recessive gene (t) or a pair of recessive alleles (tt)

Checker Board Punnett Square

- It is a square (Punnett; 1927) divided into smaller squares, which shows the mathematical (probable) result of a cross, both phenotypic and genotypic.
- It is of three types, i.e., gametic, phenotypic and genotypic. Forked line or branching system is also used to know phenotypic and genotypic probabilities.

Wild and Mutant Alleles

- It is the character, which was originally present in the population and is usually most common and dominant, whereas recessive allele is less common and forms from wild allele through mutation. It is also called mutant allele.

Homozygote (Homozygous Individual)

- Diploid cell or organism having two identical alleles of a specified gene. E.g., TT and tt (homozygous dominant and homozygous recessive).

Heterozygote (Heterozygous Individuals)

- Diploid cell or individual having two different alleles of a specified gene. E.g., Tt. The terms 'homozygote' and 'heterozygote' were coined by Bateson and Saunders (1902)

12.5. DIFFERENCES BETWEEN HOMOZYGOUS AND HETEROZYGOUS INDIVIDUAL

Homozygous Individual	Heterozygous Individual
1. It is pure for a trait and gives rise homozygous individual.	1. It is impure and produces offspring with different genotypes by selfing.
2. The alleles are similar.	2. The alleles are dissimilar.
3. Homozygous individual can carry either domination or recessive alleles but not both.	3. Heterozygous individual has both dominant and recessive alleles.
4. It produces one type of gametes and not shows extra vigour.	4. It produces two types of gametes and individual shows extra vigor called hybrid vigour or heterosis.

Hybrid

- It is an organism produced after crossing two genetically different individuals, such process of obtaining hybrids is known as hybridisation.

Genotype

- It is the genetic constitution of an individual cell or organism, e.g., genotype of hybrid tall is Tt. Pure tall is TT and dwarf is it.

Phenotype

- It represents the observable characters of an organism, which develop as a result of interaction between genotype and environment. E.g., phenotypic tall pea plant can be genetically TT or Tt.

12.6. DIFFERENCES BETWEEN PHENOTYPE AND GENOTYPE:

Phenotype	Genotype
1. The characters of an organism, which are visible from outside are called phenotypic characters or phenotype.	1. Genotype is the sum total of heredity of genetic make up.
2. The phenotype. Ratio of F ₂ – generation in Mendel's monohybrid cross is 3:1.	2. The genotype ration of F ₂ – genotype in Mendel's monohybrid cross is 1:2:1.
3. The phenotype of the pure tall and hybrid tall is same (tall).	3. The genotype of the pure tall is TT and that of the hybrid is Tt.

Gene Pool

- It is the aggregate of all genes and their alleles present in an interbreeding population.

Reciprocal Cross

- It is a cross involving two types of individuals, where the male of one type is crossed with female of the second type and vice-versa.

Back Cross

- It is performed between F1 hybrid and any one of its parents Test Cross.
- When F1 hybrid is crossed with its recessive parent.
- It is called as test cross. By test cross the heterozygosity and homozygosity of the organism can be tested.
- The offspring will be 100% dominant (Tt) if the individual, which crossed with recessive parent (tt), was homozygous dominant, (TT) and in case of hybrid individual. (Tt) the ratio will be 50% dominant and 50% recessive.
- The test cross ratio in monohybrid cross is 1:1 and in dihybrid cross, ratio will be 1:1:1:1.

Monohybrid Cross

- It is made to study the inheritance of a single pair of allele. Monohybrid Ratio is obtained in F2 –generation, when a monohybrid cross is made and the offspring's, of F1-generation are self-breed.
- Phenotypic ratio of monohybrid cross is 3:1 and genotypic ratio is 1:2:1 (RR:Rr:rr).

Dihybrid Cross

- It is made to study the inheritance of two pairs of factors or alleles of two genes.

Tri Hybrid Ratio

- It is the ratio obtained in F2 generation raised from a trihybrid cross followed by selfing or inbreeding of F1 individuals. Eight phenotypes are formed.
- The phenotypic ratio is 27:9:9:9:3:3:3:1.

12.7. DIFFERENCES BETWEEN MONOHYBRID AND DIHYBRID CROSS

Monohybrid Cross	Dihybrid Cross
1. When two plants or animals with a pair of contrasting characters are crossed with each other. It is called monohybrid cross.	1. When two plants or animals with two pairs of contrasting characters are crossed with each other. It is called dihybrid cross.
2. It produces genotypic ratio as 1:2:1 in F ₂ -generation.	2. It produces genotypic ratio of 1:2:1:2:4:2:1:2:1.
3. The phenotypic ratio of F ₂ -generation was 3:1	3. The phenotypic ratio of F ₂ -generation was 9:3:3:1.

Pure Line

- It is strain of genetically pure, true breeding individuals, which have been derived from a single self-fertilised homozygous ancestor.

Genome

- It is the complete set of chromosome where every gene chromosome is represented singly as in gamete.
- A single genome is present in haploid cells, two in diploid cells and many in polyploidy cells.

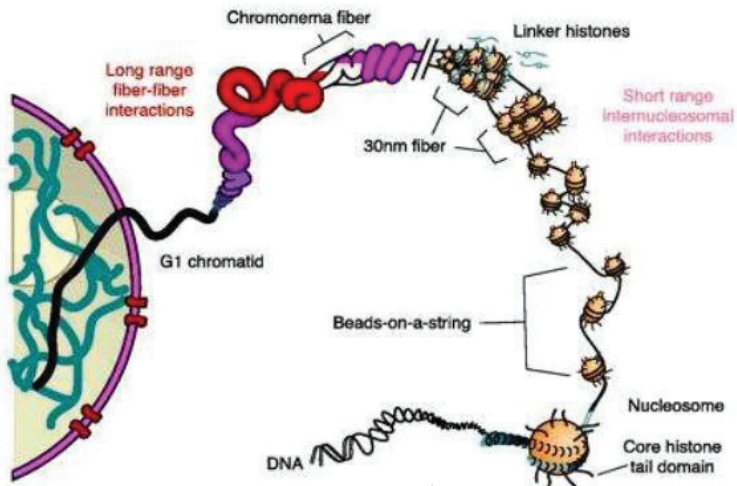


FIGURE 36: GENOME ORGANIZATION IN EUKARYOTES

Chapter 13

Multiple Allelism and Inheritance of Blood Group

ABO BLOOD GROUP IN HUMAN:

- Some traits have genes with more than two alleles. These are called multiple alleles.
- The human blood group gene provides an example of multiple allele as well as an interesting dominance relationship.
- There are three A,B,O blood group alleles, usually given the symbols I^A , I^B and I^O . I^A and I^B are codominant to each other, but are both dominant to I^O .

13.1. INHERITANCE OF BLOOD GROUPS:

- Blood group A, B and O were discovered by Karl Landsteiner, whereas blood group AB was discovered by Von Decastello and Sturli.
- A set of three multiple alleles present on the chromosome are responsible for four types of blood groups.
- Blood group AB is called universal recipient blood group because it contains antigen A and antigen B but none antibody, therefore people of blood group AB can receive transfusions of blood from people of any of the four blood groups (i.e., A, B, AB or O) but can donate only to AB blood group individual.
- Blood group O is called universal donor because this blood group have neither A nor B antigens, so their blood can be transfused into any person (i.e. person of blood) group A, B, AB or O).

13.2 .SOME OTHER EXAMPLES OF MULTIPLE ALLELISM

- Color loci in corn
- Skin color in rodents
- Eye color in *Drosophila*.
- Self-incompatibility genes in some plants.

The characteristic features of multiple alleles are that they are usually located on the same locus and hence, there is absence of crossing over between them.

Multiple alleles can be found only in population and in any diploid individual, only two alleles can be found.

Rh Factor

- It was discovered by Landsteiner and Wiener (1940) in RBCs of Rhesus monkey (*Macaca rhesus*)
- Person with Rh factor is called Rh⁺ (Rh positive), Rh⁺ blood group is found in about 85% people and person without Rh factor is called Rh⁻ (Rh negative), such type of blood group is found in about 15% people.
- Erythroblastosis foetalis or Haemolytic Disease of the Newborn (HDN) occurs when the mother is Rh⁻ and father as well as foetus is Rh⁺.

Pleiotropy

- When a gene affects many aspects of phenotype or controls several phenotypic characters, is called pleiotropic gene and this phenomenon is called pleiotropy.

13.3. SOME EXAMPLES OF PLEIOTROPY

- ‘Vestigial Wing’ mutation of *Drosophila* not only controls the size and shape of the wings but also affects several other features including reduced fecundity.
- Gene for phenylketonuria in human beings also interferes with synthesis of melanin pigment.
- Sickle-cell anaemia.
- Pleiotropic genes along with their main effect may also act as modifiers for different genes and thus, have more than one effect.

Chapter 14

Polygenic Inheritance

POLYGENIC INHERITANCE:

- Genes that acting individually have a small effect but that collectively produce a significant phenotypic expression are called polygene, e.g., genes for height or weight.
- Polygenes show polygenic inheritance or quantitative inheritance. As genes are present on different chromosomes, they will segregate independently during meiosis.
- That's way polygenic inheritance becomes a genetic problem for dihybrid cross (i.e., for two genes) and possibly much more, if more than two genes are involved for a particular trait.
- Inheritance of kernel color in wheat (H Nilsson-Rhrl; 1909) and human skin color (Davenport; 1910) are important examples of polygenic inheritance.

14.1 TYPES OF GENE INTERACTION

1. Intragenic
2. Intergenic

- In intragenic, two alleles of a gene, which are present on same gene locus on the two homologous chromosomes react to produce modified phenotype. E.g., incomplete dominance, codominance, multiple allele and lethal genes, etc.
- Intergenic interaction is one where two or more independent genes belonging to same or different chromosomes interact to form different expression. E.g., epistasis, duplicate genes, complementary genes, supplementary genes, inhibitory genes, etc.

CHROMOSOME THEORY OF INHERITANCE

- It was proposed by Walter Sutton and Theodor Boveri in 1902.
- Sutton and Boveri made a correlation between Mendel's conclusions about genes (inherited traits) and the behavior of chromosomes during mitosis and meiosis.

- Chromosomes are in pairs and genes or their alleles are located on chromosomes.
- Homologous chromosomes separate during meiosis so that alleles are segregated.
- Meiotic products have one of each homologous chromosomes but not both.
- Fertilization restores that pairs of chromosomes.

14.2 ESSENTIAL FEATURES OF INHERITANCE

- Gene is a unit of inheritance.
- Each gene has its own locus.
- Each gene has two or more alleles.
- Each character is governed by a specific gene.
- Alleles of a gene govern the alternative forms of the same trait.
- Each somatic cell have two copies of each gene.
- Each gamete have only one copy of each gene.
- Both male and female parents contribute equally to the progeny.

14.3. INTERACTION OF GENES

- Mendel considered that each characters controlled by single gene, but we know today that a character may be controlled by two or more than two genes.
- In such a situation, two or more than two genes may interact to give rise to a particular phenotype.
- If a and B are two genes, both dominant over their respective recessive alleles a and b, then the interaction will depend upon
 - The presence of both dominant alleles A and B
 - The absence of A
 - The absence of B
 - The absence of both A and B.
- Interaction of genes resulted into abbreviated phenotypic ration, i.e., the phenotypic ratio other than expected 9:3:3:1.

14.4. INTERACTION OF FACTORS AND THEIR MODIFIED RATIO

Interaction of Gene	Modified F2 Ratio
1. Interaction of complementary genes	9:7
2. Interaction of supplementary genes	9:3:4
3. Interaction of epistatic genes	12:3:1
4. Interaction of inhibitory genes	13:3
5. Interaction of duplicate genes	15:1
6. Interaction of polymeric genes	9:6:1

1. Epistasis

- When a gene at one locus 'Masks' or inhibits the expression of a gene at a distinct locus, such an interaction is called Epistasis.
- Epistatic interaction means that one gene masks or changes the effect of another gene.
- Epistasis is the interaction between different genes (non-alleles), whereas dominance is the interaction between different alleles of the same gene.
- Dominance works at interallelic but intragenic level, whereas epistasis works at intergenic level, whereas Epistasis works at intergenic level.
- In Epistasis, out of two independent genes one express itself called Epistatic, while other is masked called hypostatic.
- Such interactions give modified F2 ratio as 12:3:1.

2. Complementary Genes

- W Bateson and RC Punnett observed complementary gene interaction in sweet pea (*Lathyrus odoratus*).
- In such type of interaction, two separate pairs of genes interact to produce the phenotype in such a way that neither of the dominant genes is expressive unless the other one is present.
- Complementary gene interaction is found in sweet pea (*Lathyrus odoratus*) and shows 9:7 in F₂-generation.

3. Supplementary Genes

- In supplementary gene interaction, two non-allelic gene pairs control the phenotypic expression of a trait, but the dominant gene of one pair expresses itself only when the dominant gene of the other pair is present, whereas the dominant gene of the other pair is expressed even in absence of the dominant gene of the first pair.
- Interaction of supplementary gene is found in flower color of Snapdragon (*Antirrhinum majus*), which shows 9:3:4 ratios in F₂-generation.

4. Duplicate Genes

- When two pairs of genes are involved and both of them are able to produce the same effect singly as well as in combination, these are called duplicate genes.
- Which shows triangular and top shaped capsule in 15:1 in F₂-generation?
- The resulting variety of possible genotypes can give a large number of phenotypes and when the influence of the environment is also considered,
- It is hardly surprising that almost every possible variety of phenotype can be observed in a large population.
- This is termed as continuous variation.

Chapter 15

Linkage and Recombination**15. LINKAGE AND RECOMBINATION:**

- Linkage is the tendency of alleles of different genes to be passed together from one generation to next.
- If the chromosomes in a gamete producing cell are not rearranged for lack recombination) during meiosis, it would be expected that all the genes on a particular chromosome.
- That is the genes are linked together on their chromosomes.
- Linkage was discovered by W. Bateson, ER Saunders and RC Punnett (1905), when they were working with sweet pea (*Lathyrus odoratus*).
- They found test cross ratio 7:1:1:7 instead of 1:1:1:1 and described it as coupling and repulsion.
- Linkage is an exception to the law of independent assortment.
- It is inversely proportional to the distance between the genes on chromosomes.
- TH Morgan (1910) reported linkage in *Drosophila* and gave the term linkage to this phenomenon.
- He also reported that coupling and repulsion are two aspects of the linkage.
- A linkage group of linked genes, i.e., all the linked genes of chromosomes form a linkage group.
- The number of linkage group corresponds to the haploid number of chromosomes.

15.1. SOME ORGANISMS AND THEIR CHROMOSOMES NUMBERS

Organism	Haploid Number of Chromosome	Linkage Group
<i>Drosophila</i>	N=4	4
Garden pea (<i>pisum sativum</i>)	N=7	7
Maize (<i>Zea mays</i>)	N=10	10
Human (<i>Homo sapiens</i>)	N=23	23

Number of linkage groups in prokaryotes such as bacteria, cyanobacteria or blue-green algae, mycoplasma, etc., is one.

15.2. TYPES OF LINKAGE:

- Complete linkage when only parental combinations are produced. Linkage is called complete linkage.
- It is exhibited when the genes for a particular character are present very close to one another.
- It is rarely found nature. Crossing over remains absent in this case, e.g., male *Drosophila*, female silkworm (*Bombyx mori*).
- Incomplete linkage when some recombinations are also produced along with parental combinations, linkage is called incomplete linkage.

15.3. LINKAGE MAPS

- The graphical representations of relative distances between linked genes of chromosome are called linkage map or genetic map.
- As recombination frequencies or cross over frequencies are directly proportional to distances between genes, hence these values are used in formation of linkage maps.
- In linkage map, one map unit is equal to 1% recombination. It is also called Centi Morgan after the name of 'Thomas H Morgan'.
- Linear order of genes on chromosome can be estimated by determining the cross-over percentage between genes.

$$\text{Frequency of crossing over} = \frac{\text{Total no. of recombinant}}{\text{Total progeny}} \times 100$$

The first chromosome maps were prepared by Sturtevant in 1911.

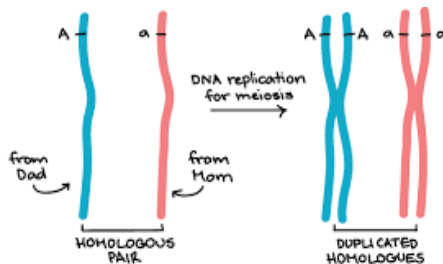


FIGURE 37: LINKAGE

Chapter 16

Crossing Over

16.1. CROSSING OVER

- It is a process occurring in the prophase-I of meiosis, in which exchange of non-sister chromatids called genetic recombination, takes place.
- The term ‘crossing over’ was introduced by Morgan and Cattell for separation of linked genes.
- The location of a gene on a chromosome is called a locus (Pl. Loci).
- The loci of the genes on a chromosome are arranged in a linear sequence.
- The two alleles of a gene in a heterozygote occupy corresponding positions in the homologous chromosomes, that is allele.
- A occupies the same position in homologue I that allele ‘a’ occupies in homologue.
- Crossing over involves the breakage of each of two homologous chromosomes (actually chromatids) and the exchange of parts.
- Crossing over occurs at polytene sub-stage of prophase-I of meiosis (at molecular level) but visible at diplotene sub-stage.
- Chiasmata are the points, where crossing over takes place.
- Crossing over is directly proportional to distance between the linked genes. i.e., more the distance between genes on the chromosome more will be the chance of crossing over.
- Crossing over occurs at pachytene stage after the synapsis of the homologous chromosomes has occurred in prophase-I of meiosis.
- Since chromosome replication occurs during S-phase, meiotic crossing over occurs in the post-replication tetrad stage.
- That is after each chromosome has doubled such that four chromatids are present for each pair of homologous chromosomes.
- Each pair of synapse homologs is called a tetrad, because it consists of four chromatids.
- Crossing over takes place between any, two non-sister chromatids of a homologous pair of chromosomes.
- Crossing over refers to the exchange of segments of chromosomes (i.e. the process), whereas recombination implies the formation of new combinations of genes (i.e., one possible outcome of process).

- The most accepted theory of crossing over is per costly theory of strain theory, given by CD Darlington by which, as a result of twisting of homologous chromosomes, a strain develops, which causes breakage of chromatids at the point of contact.
- The broken segments unite with other segments to bring about crossing over.

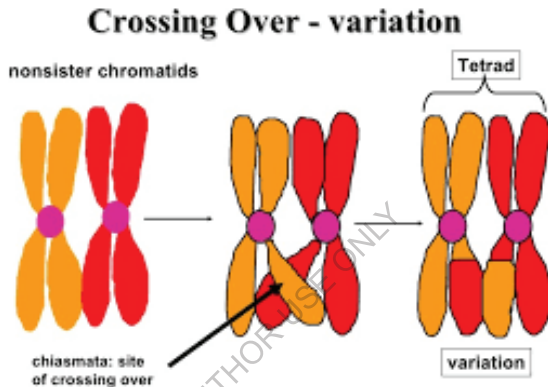


FIGURE38: CROSSING OVER

16.2. SIGNIFICANCE OF CROSSING OVER

- It provides proof for linear arrangements of genes on chromosomes.
- It is the basis for linkage maps or genetic maps construction.
- Recombination of new gene combinations are produced due to crossing over, which is a source of variations.

16.3. MUTATION

- It is a phenomenon, which results in alteration of DNA sequences and consequently results in changes in the genotype and the phenotype of an organisms.
- This phenomenon also leads to variation in DNA.
- Genes are located on chromosomes, therefore any alteration in chromosomes results in abnormalities or aberrations.
- These aberrations are commonly seen in cancer cells.

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- Mutations also occur dna single base pair of DNA. This is called point mutation, .g., sickle cell anaemia.
- Deletion and insertion of base pairs o DNA, causes frame-shift mutations.
- The mutagens, which mutations are many physical, chemical factors and UV radiations.
- The term mutation was coined by Hugo de Vries (1901).
- Somatic variation is also called the modification or acquired character because they got by individual during its life time.
- The frequency of crossing over value (COV) is calculated using formula

$$\frac{\text{Total no. of recombinant}}{\text{Total progeny}} \times 100$$

- Janessens (1909) was the first person to discover chiasma formation and related processes of crossing over.



FIGURE 39: MUTATION IN PLANTS

Chapter 17

Sex-Determination

17. SEX DETERMINATION:

- In Dioecious species (separate sexes), there are several means to determine sex.
- The chromosomes involved in sex determination are called sex chromosomes.
- All other chromosomes are called Autosomal chromosome Autosomes. Although sex chromosomes provide the most common means of sex determination.

17.1. SOME EXAMPLES OF SEX DETERMINATION:

1. In bees, males are haploid (n), while females are diploid ($2n$).
2. Sex may be determined by a single allele or multiple alleles as in some wasps.
3. By Environmental factors as in some turtles, these have indeterminate genetic sex-determining mechanisms.
4. The temperature at which the eggs are incubated determines the sex of the turtles.
5. In some species, warm nests yield mostly males and cool nests yield mostly females. In other species of turtles this is reversed.

17.2.FIVE BASIC TYPES OF CHROMOSOMAL MECHANISMS

- In human XX-YY system, in which females are homomorphic XX and males are heteromorphic XY.
- This is found in mammals including humans and some insects including *Drosophila*.
- In humans, females have 23 homomorphic pairs and males have 22 homomorphic pairs a heteromorphic pair.
- During meiosis, females produce only one kind of gametes all having one X-chromosome.
- Males produce two kinds of gametes, one with an X and the other with a Y-chromosome. Females are homogametic and males are heterogametic.

- In birds ZW-ZZ system, in which females are heteromorphic ZW and males are homomorphic ZZ.
- This occurs in birds, some fishes and moths. It is essentially the opposite of XY in mammals.
- In insects XX-XO system, in which females have 2X-chromosomes. Males have only one X and no additional sex chromosomes. Females have an even number of chromosomes and males have an odd number of chromosomes.
- This occurs in many species of insects. This was the first sex determining chromosome was named the X in 1905.
- Gametes of males have either an X-chromosome or no sex chromosome.
- ZO-ZZ type This type of sex determination occurs in some butterflies and moths.
- It is exactly opposite to the XX-XO type of sex determination. Here the females have odd sex chromosome (AA+Z), while males have two homomorphic sex chromosomes (AA+ZZ).
- Compound chromosome system can be very complex with multiple numbers of X and Y-chromosomes, e.g., in *Ascaris incurva*, a nematode, there are 26 autosomes, 8 X-chromosomes and Y-chromosomes.
- Male have 26A+8X+Y for 35 chromosomes. Females have 26A+16X for 42 chromosomes. This type of system is also common in spiders.
- Sex determination in the XY system is the most studied because it is found in humans and *Drosophila*.
- Does the X-chromosome or the Y-chromosome determine the sex? It varies from species to species.
- In *Drosophila*, the greater the number of X-chromosomes relative to the autosomes, the more likely the individual will be female.
- Sex balance theory or genic balance theory states that the X-chromosome determines the sex of the individual and that sex is a dosage phenomenon, where the ratio of the amount of the X relative to the autosomes determines the sex.
- In addition, environmental effects can influence the development of the intersex flies.

Different Phenotypes, Chromosomal and X/A Ratio in Drosophila:

Phenotype	Chromosomal Complement	X/Autosomal Set
Normal female	XX+2n autosomes	1.00
Normal male	XY+2n autosomes	0.50
Metafemale	XXX+2n autosomes	1.50
Metamale	X+3 n autosomes	0.33
Intersex	XX+2n autosomes	0.67

- Genic balance theory was originally proposed by Calvin Bridges in 1921.
- Gynandromorph is a dioecious organism with sexual mosaic.
- Gynandromorph is typically male in certain portions of the body and typically female in others, it means body of gynandromorphy is half male and half female.
- This term was coined by Goldschmidt (1915). Gynandromorphism has been reported in *Drosophila*, human silkworms, bees, butterflies, beetles, etc.,

17.3. SEX-LINKED INHERITANCE:

- The characters of traits carried on sex chromosomes are called sex-linked traits and their inheritance is called sex-linked inheritance.
- First of all TH Morgan (1910) observed sex-linked inheritance in *Drosophila* (fruit fly).
- In humans, colour blindness (red-green blindness) and haemophilia (Bleeder's disease) are important sex-linked (X-linked) diseases.
- The genes located exclusively on X-chromosomes are called X-linked and the genes that occur only on Y-chromosomes are called holandric genes, whereas the genes present on homologous regions in sex chromosomes are called XY-linked genes.
- The inheritance, i.e., male transmits his X-linked recessive genes to his grandson through his daughter.
- As males have only one X-chromosome (XY – sex chromosome) in contrary to female, which have two X-chromosomes, male is called hemizygous and can have only one allele at a locus.
- Because of this, one copy of a recessive allele will be expressed in the phenotype in male.

A. Autosomal Dominant Inheritance

- It is caused by dominant autosomal gene.
- In autosomal dominant inheritance if the trait is rare, most matings are between heterozygotes (affected) and homozygous (unaffected) recessive, which lead to predominantly 1:1 phenotypic ratios for all children, regardless of sex.
- The dominant phenotype should appear in every generation. E.g., Achondroplasia, polydactyly, brachydactyly, Huntington's chorea, phenylthiocarbamide testing, aniridia etc.,

B. Autosomal Recessive Inheritance

- It is caused by recessive autosomal gene present in homologous condition both parents.
- Are affected, all children will be affected. Often found in consanguineous marriages. Most affected children have normal parents, e.g., alkaptonuria, albinism, Tay-sach's disease, gaucher's disease, sickle cell anemia, Thalassemia, and cystic fibrosis etc.

17.4. SEX LINKED DOMINANCE:

The two sex-linked as dominance and recessive are given below

1. Sex-Linked Dominance

- No generations are skipped. Affected males must come from affected mothers.
- All the daughters, but none of the sons, of an affected father are affected. Approximately half of an affected female's sons and daughters are affected.

2. Sex – Linked Recessive

- Males are most affected. Affected males have carrier mother, who are known to have affected brothers, father or maternal uncles.
- Affected females come from affected father and carrier mother. Affected female's sons must be affected.
- Approximately half the sons of carrier females should be affected.

3. Sex-Limited Traits

- They are traits that are autosomally inherited and they are expressed in one sex but not in the other.

- Some examples include sexually dimorphic plumage in birds, milk yield in mammals, antlers in deer, and baldness in humans (males).
4. Sex-Influenced Traits
- They appear in both sexes but more so in one sex than another. Male pattern baldness in humans is an example.
 - The male hormone testosterone is needed for full expression of baldness.
 - Because of this hormone difference, the allele for baldness behaves as a dominant trait in males (expressed when heterozygous) but behaves as a recessive allele in females (must be homozygous to be expressed).
5. Pedigree Analysis
- The analysis of traits in several generations of a family is called pedigree analysis.
 - In the pedigree analysis, the inheritance of a particular trait is represented in the family tree over generations.
 - It is a strong tool to trace the inheritance of a specific trait, abnormality or disease.

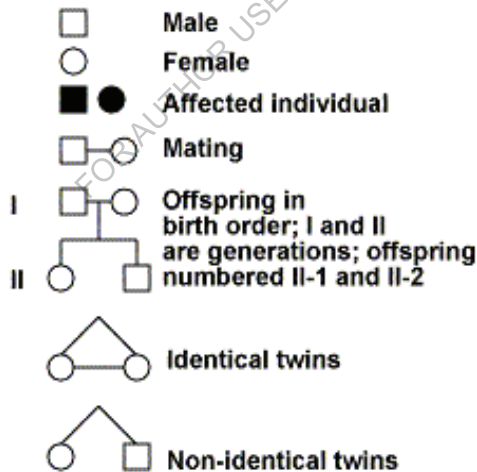


FIGURE 40 : PEDIGREE SYMBOLS

Chapter 18

Genetic Disorders

18. MENDELIAN DISORDERS:

- These disorders occur mainly due to alteration or mutation in a single gene.
- These disorders may be dominant or recessive.
- These are transmitted from one generation to the next following Mendel's principles of heredity.

18.1. EXAMPLES OF MENDELIAN DISORDERS

1. Haemophilia
2. Colour blindness
3. Sickle-cell anaemia
4. Phenylketonuria
5. Thalassemia, etc.

1. Hemophilia

- Haemophilia (bleeder's disease) is X-linked recessive disease, which was first studied by John Cotton in 1803. Haemophilia was common in royal families of Europe. Haldane discovered pedigree of haemophilia in royal family, which started from Queen Victoria due to mutation in gametes.
- Haemophilia-A characterised by lack of antihemophilic globulin (Factor-VIII). About 4/5 cases of haemophilia belong to haemophilia-A.
- Haemophilia-B or Christmas disease results from a defect in Plasma Thromboplastic Component (PTC or factor-IX).
- As hemophilia is caused by recessive X-linked gene, a woman, which carry the gene of that disease is called carrier (i.e., she is carrier but non-sufferer and would transmit it to 50% of her sons, even if the father is normal).

2. Color Blindness

- It is a condition, in which certain colours (green or red) cannot be distinguished, due to lack of one or more colour absorbing pigments in the cone cells of retina.
- In humans, the most common colour blindness is red-green colour blindness, which is a sex linked (i.e., X-linked recessive) defect caused by a recessive gene and is thus more common in males than females.
- Pedigree of colour blindness was first described by Horner (1876). Red-green colour blindness, in which the green-sensitive cones in the retina are defective, is due to an X-linked recessive allele.

The symbols needed to follow this allele through a Punnett square are

1. X = X-chromosome carrying the normal (dominant)
2. X^c = X-chromosome carrying the defective (recessive) allele.
3. Y = Y-chromosome, with no allele for this gene on it.

3. Sickle-Cell Anemia

- This is an autosome-linked recessive trait that can be transmitted from parents to the offspring when both the partners are carrier or affected for the gene.
- The disease is controlled by a single pair of allele. HbA and HbS , out of the possible genotypes only homozygous individuals for HbS ($HbSHbS$) show the diseased phenotype.
- Heterozygous ($HbAHbS$) individuals appear apparently unaffected but they are carrier of the disease as there is 50% probability of transmission sickle-cell trait.
- The mutant haemoglobin molecule undergoes polymerisation under low oxygen tension causing the change in the shape of the RBC from biconcave disc to elongated sickle like structure.

4. Phenylketonuria

- This is also an Autosomal recessive trait. The affected individual lacks an enzyme that converts the amino acid phenylalanine into tyrosine.
- This phenylalanine is accumulated and converted into phenyl pyruvic acid and other derivatives.
- These are excreted through urine because of its poor absorption by kidney.

5. Thalassemia

- It is an inherited Autosomal co-dominant blood disease. In thalassemia, the genetic defect results in reduced rate of synthesis of one of the globins chains that make up hemoglobin.
- Reduced synthesis of one of the globins chains can cause the formation of abnormal hemoglobin molecules, thus causing anemia.

18.2. CHROMOSOMAL DISORDERS:

- The chromosomal disorders are caused due to absence or excess or abnormal arrangement of one or more chromosomes.
- The failure of segregation of chromatids during cell division cycle results in the gain or loss of chromosomes called Aneuploidy.
- The failure of cytokinesis after telophase stage of cell division results in an increase in a whole set of chromosomes in an organisms.
- This phenomenon is called polyploidy. The condition is commonly seen in plants.
- Sometimes, either an additional copy of a chromosome may be included in a person or may lack one of one pair of chromosomes.
- These situations are called as trisomy or monosomy of a chromosome, respectively.

1. Down's Syndrome

- This genetic disorder is also caused due to the presence of an additional copy of X-chromosome resulting into a karyotype of 47 (XXY).
- Such individuals have overall masculine development, however, the feminine development (development of breast, i.e., gynaecomastia) is also expressed. Such individuals are sterile.

2. Turner's Syndrome

- This disorder is caused due to the absence of one of the X-chromosomes. i.e., 45 with XO.
- These females are sterile as ovaries are rudimentary besides other features including lack of other secondary sexual characters.

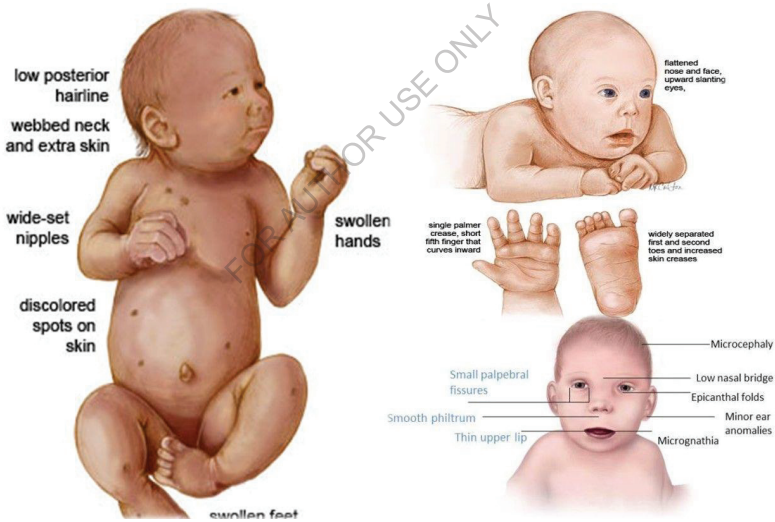
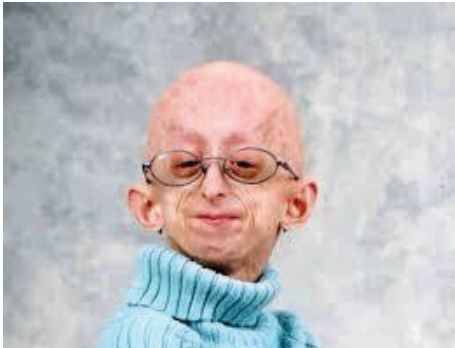


FIGURE 41: GENETIC DISORDER CHILDS

Chapter 19

General Laboratory Safety Precaution

19.1. DO AND DON'T LABORATORY SAFETY:

Safety in the Laboratory Should always be in your mind throughout this manual. Safety recommendations are given, below are some general considerations that anyone in a laboratory should know.

- Follow all instructions carefully. Use special care when you see the word caution in your laboratory instructions. Follow the safety instructions given by your teacher.
- Determine the location of Fire Extinguishers, Chemical safety showers and Eye washers, Chemical Spill Kits and alternative exit routes for lab evacuation.
- Remember that smoking, eating or drinking in the lab room is totally prohibited.
- Wear lab aprons when working with chemicals, hot material or preserved specimens.
- Wear safety goggles when using dangerous chemicals, hot liquids or burners.
- Any chemicals spilled on the hands or other parts of the skin should be washed off immediately with a plenty of running water.
- If you have an open skin wound, be sure that it is covered with a water proof bandage.
- Never work alone in the laboratory.
- Keep your work area clean and dry.
- Turn off all electrical equipment, water and gas when it is not in use, especially at the end of the laboratory period.
- Tie back long hair.
- Report all chemicals spills or fluids to your instructor immediately for proper clean up.

19.2. SPECIAL PRECAUTIONS FOR WORKING WITH HEAT OR FIRE

- Never leave a lighted Bunsen burner or hot object unattended. When an object is removed from the heat & left to cool, it should be placed where it is shielded from contact.
- Inflammable liquid bottles should not be left open, not dispensed near a naked flame, hot electric element or electric motor.
- Use test tube holders to handle hot laboratory equipments.
- When you are heating something in a container such as a test tube, always point the open end of the container away from yourself & others.
- Use only Pyrex glassware's for heating.
- Allow hot materials to cool before moving them from your lab station.
- Make sure that Bunsen burner hoses fit tightly.

19.3. SPECIAL PRECAUTIONS FOR WORKING WITH CHEMICALS

1. Never taste or touch substances in the laboratory without specific instructions.
2. Never smell substances in the laboratory without specific instructions.
3. Use materials only from containers that are properly labeled.
4. Wash your hand after working with chemicals.
5. Do not add water to acid. Instead, dilute the acid by adding it to water.
6. Mix heat generating chemicals slowly.

19.4. SPECIAL PRECAUTIONS FOR WORKING WITH ELECTRICAL EQUIPMENT:

1. Make sure the area under & around the electrical equipment is dry.
2. Never touch electrical equipment with wet hands.
3. Make sure the area surrounding the electrical equipment is free of flammable materials.
4. Turn off all power switches before plugging an appliance into an outlet.

19.5. SPECIAL PRECAUTION FOR WORKING WITH GLASSWARE'S AND OTHER LABORATORY EQUIPMENTS:

1. Become familiar with the names and appearance of all the laboratory equipments you will use.
2. Never use broken or chipped glassware.
3. Make sure that all glassware's are clean before you using it.
4. Do not pick up broken glass with your bare hands. Use a pan and a brush.
5. If a Mercury thermometer breaks, do not touch the mercury. Notify your teacher immediately.
6. Do not aim the mirror of your microscope directly at the sun.
7. Direct sun light can damage the eyes.
8. Use care handling all sharp equipments, such as scalpels and dissecting needles.

19.6. SPECIAL PRECAUTIONS FOR WORKING WITH LIVE OR PRESERVED SPECIMENS:

1. If live animals are used treat them gently. Follow instructions for their proper care.
2. Always wash your hands after working with live or preserved organisms.
3. Specimens for dissection should be properly mounted and supported. Do not try to cut a specimen while holding it in the air.
4. Do not open Petri dishes containing live cultures unless you are directed to do so.
5. Detergents (Detol 5 – 10%) should be used to sterilize and clean benches, glassware and equipment.
6. Safety cabinet should be used while working with microbes.
7. Lab coats should be worn during the work in the lab.
8. Disposable items should be collected and autoclaved.

Chapter 20

FIRST AID AND SAFETY MEASURES

20.1. FIRST AID AND SAFETY MEASURES

1. Injuries: bleeding should be reduced using bandages; the wound
2. Should be cleaned with iodine alcohol mixture and wrapped with sterile bandage.
3. Acid and fire burns: body burns must be washed immediately with tap water. Eye burns must be washed using eye washer, special cream for burns can be used.
4. Poisoning: if any toxic chemical is swallowed, the mouth must be sensed with water, in case of acid, milk is drunk, in case of alkaline, diluted acetic acid (vinegar) can be used.
5. Skin contamination requires washing with water and removal of contaminated clothing, if the contaminant is insoluble in water remove with soap and water.



FIGURE 42: FIRST AID KIT

20.2. RESPONSIBILITY

- The laboratory instructor and laboratory aides are responsible for maintaining this procedure.

20.3. PROCEDURE

- Laboratory Procedures All employees working within the District laboratories are provided with a copy of applicable Laboratory Safety System procedures (chemical Hygiene Plan) and provided with training necessary for safe operations.
- Work study students are also provided with safety training.
- All employees are required to familiarize themselves with the procedures and sign a statement acknowledging receipt and understanding of the procedures.
- The signed statement is kept on file by the Chemical Hygiene Officer.
- Laboratories are used for laboratory activities ONLY.
- If you make a mess, you are responsible for ensuring it is cleaned up.
- Disinfectant is used to wipe lab working areas before and after lab activity.
- Good laboratory practices are to be followed at all times. If a student is in doubt, direction of the instructor or laboratory aide should be sought before proceeding in ANY activity.
- All laboratory assignments MUST take place and be completed in the laboratory. NO lab activity is allowed in classrooms.
- All labs must be completed on the assigned date, no open labs are permissible unless an instructor or lab aide is present.
- Students must not be allowed to congregate in the halls outside the lab while waiting for practical exams to start. This is a fire hazard and disruptive to other classes. Give students a definite time to come to the lab to begin the practical.
- Personal Protective Equipment Safety glasses MUST be worn during ALL laboratory activity involving chemicals, boiling water, use of the Bunsen burner and dissecting by students, instructors, and laboratory aides.
- Other personal protective equipment (i.e. gloves) is to be worn as directed by laboratory setup instructions or the instructor.

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- Failure to wear appropriate personal protective equipment in laboratory situations will lead to disciplinary measures, up to and including dismissal from employment or expulsion from the college program.
- Personnel working or using the lab must confine long hair, and wear regular length lab coats. When applicable, wear gloves to protect hands from contamination. Inspect gloves before use.
- NO clogs, flip-flops, open-toed shoes, sling back heels, sandals, or bare feet are allowed in lab. Regular length lab coats must be worn at all times over all clothing.
- Know the location of all fire and first aid equipment.
- Use of the Lab Students is NEVER allowed in the laboratory areas without supervision of the instructor or laboratory aide.
- Coats, purses, extra books, or any other materials not being used in the current laboratory activity are NOT allowed in the working area. These items are to be placed in the designated area during laboratory times.
- The doors to the laboratory and chemical storage areas MUST be locked upon leaving by the instructor and/or laboratory aide.
- Damaged or Defective Equipment all equipment and materials must be kept in its designated storage area when not in use. Inspect equipment and materials before use. NEVER use damaged or defective equipment.
- If equipment is believed to be damaged or defective: • Discard in the appropriate waste receptacle, Remove from the work area and place a written notice on it of suspected damage or defect to prevent possible use, AND • Report damaged or defective equipment to the laboratory aide .
- NO food, drink, smoking, or children are allowed in the college laboratory areas at ANY time.
- NEVER walk away from Bunsen burners when in use.
- BE ALERT for unusual or unsafe circumstances in the laboratory areas. REPORT all concerns IMMEDIATELY to the laboratory aide. If you make a mess, you are responsible for ensuring it is cleaned.

20.4. HANDLING EQUIPMENT AND MATERIALS

- Properly label all glassware and slides.
- Keep all equipment and materials away from the edge of the lab benches.
- Use protective aids (e.g. test-tube clamps, hot pads) when handling hot glassware.
- Light gas burners only as instructed. Be sure no volatile or flammable materials are being used nearby. Remember to have long hair tied back, no dangling jewelry or loose clothing
- When heating a test tube do not look into the mouth of the tube or point it into the direction of any person.
- Heat sources should only be on when in use. Do not leave heat sources unattended. When in doubt, students should assume a heat source is on or hot.
- Operate electrical equipment in a dry area with dry hands.
- When removing an electrical plug from its socket, pull the plug, not the cord.
- Inform your instructor immediately if equipment or materials are broken or not working properly.
- Never return reagents back into their original containers. Keep containers capped when not in use.

20.5. HANDLING HAZARDOUS MATERIAL

- Wear appropriate eye protection as directed by the instructor.
- Wear disposable gloves when handling hazardous materials including chemicals, organisms, or bodily fluids.
- Wear protective clothing when required by the instructor.
- Follow correct disposal procedures as detailed in “Cleanup and Disposal.”
- To prevent contamination by needle stick injuries, always use new needles and lancets.
- Use the fume hood whenever noxious, corrosive or toxic fumes are produced or released.

20.6. BACTERIA AND FUNGI

1. Never open Petri dishes containing bacterial or fungal growth unless directed to do so by the instructor.
2. Dispose of all discarded bacterial and fungal cultures as directed by the instructor.

20.7. CLEANUP AND DISPOSAL

Follow correct disposal procedures – dispose of used materials promptly and in the appropriate disposal containers.

- a. Dispose of broken glass in labeled glass disposal containers.
 - b. Dispose of needles and lancets in puncture-proof sharp containers
 - c. Dispose of used chemicals in labeled and capped waste containers as directed by your instructor.
 - d. For disposal of items contaminated with organisms or bodily fluids,
 - Decontaminate slides, spectrophotometer tubes, and other glassware in 10% bleach
 - Dispose of soft items (i.e. gauze or swabs) in labeled biohazard containers.
 - Dispose of all animal tissues in labeled biohazard containers.
1. Place micropipette tips, sample vials, and contaminated glassware in a labeled waste container or designated area.
 2. See instructor for disposal of cadaver tissues.

BIOLOGY LABORATORY SAFETY RULES AND RECOMMENDATIONS

(For all students, faculty and staff using the Biology Labs)



1. **No food or drink** is allowed in any biology lab. No gum chewing. No putting on of makeup. Lunches, thermoses, and water bottles can be stored inside backpacks or purses. They are not allowed on lab benches, the floor, near computers, or in sinks. Coffee cups are not allowed in the teaching and research labs.

2. **Be polite**- no pushing, shoving, fighting or horseplay.

3. **Always wear goggles** when you work with chemicals, hot liquids, or other materials that could harm you. Use caution when working with organic solvents if you wear contact lenses (especially soft lenses) – if you're in doubt about what chemicals you're working with ask the instructor.

4. **Wear gloves** while working in the laboratory – gloves are not provided- ***you must purchase examination gloves yourself***. Gloves are available for purchase at the campus bookstore. Please let us know if you have an allergy to latex.

5. If you have allergies or asthma or think you might be pregnant please inform the instructor.

6. **Always wear** closed toed shoes and long pants or a long skirt during labs to protect your legs and feet from spills.
7. **Always alert your teacher** to any spills on your skin or clothing.
8. **Never taste** or smell ANY chemicals unless your teacher tells you it is safe to do so.
9. **Tie or pin** long hair, scarves and headwear out of the way to avoid contact with flames or chemicals.
10. **Be sure** you know how to use equipment before beginning an experiment. If you are unsure - ASK !
11. **Children** are not allowed in the laboratory. There are too many hidden dangers for young people.
12. **Mouth pipetting or drinking of solutions** is strictly prohibited.
13. **Broken glassware** or sharp metal pieces should be placed in “sharps” boxes (also labeled “broken glass”) unless it is contaminated with body fluids or microorganisms. Contaminated sharps should be placed in an appropriate labeled metal or glass container for sterilizing.
14. **Organic fluids** (e.g., ether, acetone, chloroform) **or** other volatile liquids should be used inside a fume hood.
15. **Chemical wastes** should never be placed in a sink drain without permission. Please consult your instructor in all cases.
16. **Media with microorganisms** (liquid or solid) should be placed in a Biohazard bag or labeled autoclavable beakers for sterilizing and disposal.
17. **All glassware** should be rinsed after use and test tubes placed upside down in the racks provided. Slides and coverslips should be discarded.
18. **Students and instructors should know the locations and operation of all safety equipment.** This includes fire extinguishers, eye washes, body showers, first aid kits, and the nearest exit from their room.

BIOLOGY SAFETY CONTRACT

- This biology course offers a variety of laboratory exercises on current concepts in biology often using state of the art scientific equipment and technology.
- Various teaching techniques and materials will be employed enhance student understanding. Numerous laboratory methods will be utilized in demonstrations and student experiments. Safety instruction will be given and safe practices will be stressed in all laboratory work.
- Students will exercise critical thinking for solving problems and interpreting laboratory results.

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